

Production Linkages and Regional Development: Evidence from Coal Mining in Germany*

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Abstract

This paper examines how production linkages shape the local impact of industrial activity. I study this mechanism in the context of coal mining in Germany by comparing the extraction of two types of coal that were mined under similar conditions but differed in their integration into local supply chains. I construct a novel county-level employment dataset for Germany spanning 1849–2024, covering the rise, peak, and decline of both coal industries. Using an event-study framework and a shift-share design, I find that hard coal mining, which produces strong upstream and downstream linkages, led to persistent increases in local manufacturing and service employment, particularly in input-supplying and coal-using industries. In contrast, lignite mining, with weaker linkages, led to local industrial decline, consistent with Dutch disease dynamics. I further show that the phase-out of hard coal caused substantial employment losses, particularly in formerly linked sectors. These results highlight the double-edged nature of production linkages: they fuel growth during extraction but can deepen decline when extraction ends.

Keywords: Production Linkages; Local Multiplier; Resource Extraction; Structural Change; Deindustrialization

JEL Codes: J21, O14, Q32, Q33, R11

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1 Introduction

Industry shocks produce divergent local outcomes, generating positive employment spillovers in some regions while causing stagnation or decline in others.¹ Understanding the mechanisms through which industry shocks affect regional economies is increasingly important, as rising regional inequality across advanced economies has fueled political discontent, weakened social cohesion, and increased interest in industrial and place-based policy.²

Natural resource extraction is a particularly salient example. It has long shaped regional development, from the coalfields of Europe to the oil towns of North America and the mining zones of Latin America and Sub-Saharan Africa. Yet its local effects are strikingly uneven: some mining regions experience sustained industrial growth, while others face deindustrialization or persistent decline.³ Despite extensive empirical and theoretical work, the mechanisms behind this divergence remain poorly understood. Why do some resource-rich regions thrive while others fall behind?

In this paper, I examine one potential mechanism behind the heterogeneous local effects of natural resource extraction: the strength of local production linkages. Production linkages refer to the extent to which activity in one sector stimulates economic activity in other sectors through input demand and output supply (Hirschman, 1958). I exploit a natural experiment in Germany (1849–2024), comparing the local consequences of extracting two types of coal that were mined under similar national institutions and at comparable scales, but differed markedly in their degree of local economic integration. Hard coal extraction required complex underground mining techniques and generates strong upstream linkages through demand for machinery, engineering, and metalworking. Its high energy density also made it a key input in a wide range of industries, creating substantial downstream linkages.⁴ In contrast, lignite is extracted through simpler surface mining and, due to its lower energy content, was largely used for electricity generation and local

¹For industry shocks and local adjustment, see e.g. Autor et al. (2016), Black et al. (2005), Feyrer et al. (2017), Greenstone et al. (2010), and Hooker and Knetter (2001).

²On the political and social consequences of regional divergence, see e.g. Autor et al. (2020) and Rodríguez-Pose (2018).

³A large empirical literature has documented both negative and positive long-run effects of resource extraction on local economic development. For negative effects, see e.g. Esposito and Abramson (2021), Papyrakis and Raveh (2014), and Shao et al. (2020); for positive effects, see e.g. Aragón and Rud (2013), Cavalcanti et al. (2019), and Michaels (2011). For overviews, see Havranek et al. (2016) and van der Ploeg (2011).

⁴*Upstream linkages* refer to input relationships where local industries supply goods or services to the mining sector. *Downstream linkages* refer to industries that use mining outputs (here, hard coal or lignite) as inputs in their own production processes.

heating. As a result, it is much less integrated into surrounding industries. The setting allows for a within-country comparison of two closely related industries that operated under the same legal frameworks, national institutions, and macroeconomic conditions but differed markedly in their degree of local economic integration. This provides a credible framework to isolate the role of production linkages in shaping local employment, both during periods of industrial expansion and in the aftermath of sectoral decline.

To derive testable predictions about how extractive industries shape regional labor markets, I adapt the local multiplier framework from Moretti (2011). Specifically, I hypothesize that sectors like hard coal mining, which are embedded in local production networks through upstream demand for inputs and downstream use in industrial processes, generate stronger spillovers to other tradable sectors and amplify demand for local services.⁵ By contrast, sectors that operate as enclaves, such as lignite mining, with limited integration into local supply chains, are less likely to produce positive spillovers and may even crowd out other tradables by raising input costs.⁶ Importantly, the very linkages that support growth during extraction may also increase exposure to decline. My framework thus implies that hard coal regions experience larger employment gains during the extraction period, but also sharp contractions during the coal phase-out. Lignite regions, by contrast, are likely to see few benefits, or even adverse effects, during extraction, but smaller losses during decline. I test these hypotheses empirically by analyzing the local effects of hard coal and lignite extraction in Germany during the coal extraction and phase-out period.

To examine whether differences in local linkage intensity shaped the long-term employment effects of resource extraction, I construct a novel county-level panel dataset for Germany from 1849 to 2024. I harmonize historical occupation titles from 28 censuses and align administrative boundaries, accounting for major political transitions from Prussia to modern-day Germany. My dataset spans more than 170 years and covers the rise, peak, and decline of both hard coal and lignite mining. To my knowledge, this is the first county-level dataset that captures local employment over such a long temporal and industrial scope in Germany. I link this dataset to a historical

⁵This also echoes insights from staple theory, which emphasizes that the long-run impact of extracting a natural resource (the *staple*), depends on the extent to which it creates linkages to other sectors of the economy (Gunton, 2003; Watkins, 1963).

⁶Higher mining employment can raise local wages, increasing production costs for all firms. Since tradable goods are sold in national or international markets, local producers cannot pass these costs on to consumers and may become uncompetitive.

input–output table to classify manufacturing sectors by their upstream and downstream linkages to coal mining, enabling a disaggregated analysis of sectoral spillovers.

To identify the local effects of coal extraction, I combine two complementary empirical strategies. First, I implement an event-study design comparing employment trends across hard coal, lignite, and non-coal regions after the onset of large-scale extraction and after the coal phase-out. To address concerns that mining activity may be endogenous to local conditions, I proxy for extraction using spatial variation in coal-bearing strata, which were formed long before modern economic activity and are plausibly exogenous (De Pleijt et al., 2020; Esposito & Abramson, 2021; Fernihough & O’Rourke, 2021).⁷ Second, I estimate a shift–share specification that interacts annual variation in global coal prices with county-level exposure to coal deposits. This design exploits plausibly exogenous variation in exposure to global shocks to estimate the cyclical effects of extraction on local labor markets.

My findings show that the strength of local production linkages plays a central role in shaping the long-term effects of resource extraction. The event-study results show that the large-scale expansion of hard coal mining, characterized by strong local linkages, led to sustained gains in manufacturing employment. On average, employment in manufacturing increased by approximately 19 percent relative to non-coal regions. These gains were driven by employment growth in linked industries, those supplying inputs to mining (upstream), those using coal as an input (downstream), and those with both types of linkages. Service employment also increased, consistent with local demand spillovers from rising income and population, while agricultural employment declined as regions industrialized. In contrast, lignite mining, marked by weaker local linkages, led to industrial decline. Manufacturing employment fell by approximately 20 percent alongside similar contractions in the service sector. These patterns suggest limited spillovers and crowding-out effects, consistent with Dutch disease dynamics (Corden & Neary, 1982; Krugman, 1987).

I use the shift–share approach to complement the event-study findings by leveraging a distinct identification strategy that captures short-run responses to global coal price fluctuations. While the estimated magnitudes are smaller, reflecting that the shift–share design identifies marginal responses to price fluctuations rather than long-term structural effects, they support the conclusion

⁷I confirm that the distances from county centroids to hard coal and lignite fields are highly correlated with the distances from county centroids to carboniferous and lignite rock strata, respectively.

that coal booms boosted labor demand in hard coal regions but not in lignite areas, reinforcing the central role of production linkages in shaping the local effects of resource extraction.

After the phase-out of hard coal beginning in the 1960s, hard coal regions experienced pronounced deindustrialization, with manufacturing employment declining by about 18 percent relative to non-coal areas. Importantly, this vulnerability was not due to a lack of industrial diversification: I show that hard coal regions became more diversified across manufacturing industries during the extraction period. Instead, my results show that it was the depth of integration with the coal economy, particularly in sectors tightly linked to mining, that shaped the severity of losses once coal was phased out. The decline of the lignite industry, which began in the 1990s, also had negative effects, but these were substantially smaller.⁸ Taken together, my results suggest that regions with a strongly integrated key industry tend to gain more during periods of industrial activity but face sharper contractions when that industry declines. In contrast, regions where a major industry is weakly linked to other local sectors tend to experience adverse effects during its expansion, but more limited losses during its decline.

Contribution to the literature. This paper makes a central contribution to understanding how local production linkages shape the long-run regional effects of industrial activity. I study this mechanism in the context of natural resource extraction by comparing two closely related industries, hard coal and lignite mining. Both operated under the same legal frameworks and were subject to the same national economic conditions. While they differed in extraction methods and industrial uses, the two sectors were otherwise very similar. This setting allows me to isolate the role of production linkages in a way that is difficult to achieve in cross-country or single-industry studies, where institutional and structural differences typically obscure the underlying mechanism. Moreover, the long-run panel that I constructed allows me to examine how production linkages shape local economic outcomes both during industrial activity and as it declines.

This paper mainly contributes to four strands of literature. **First**, it extends the literature on local multipliers by demonstrating that both the size and direction of industrial spillovers to tradable and non-tradable sectors can vary substantially, even within a single industry, depending on the intensity of local production linkages (Greenstone et al., 2010; Moretti, 2011). In doing

⁸While lignite extraction in Germany has not yet been fully phased out, production has declined substantially since the 1990s.

so, my paper also informs debates on place-based and industrial policy by identifying the types of sectors most likely to generate large local multipliers. My results suggest that policies targeting sectors with strong local linkages are more likely to generate durable local gains. In contrast, support for enclave-type industries with limited economic integration may even crowd out other sectors and constrain broader development (Austin et al., 2018; Criscuolo et al., 2019; Juhász et al., 2024; Kline & Moretti, 2014; Lane, 2025; Moretti & Thulin, 2013; Neumark & Simpson, 2015; Sheremirov & Spirovska, 2022).

Second, it contributes to research on structural change and regional resilience by showing that even relatively diversified regional economies can become highly vulnerable when their lead sectors are tightly embedded in local input–output structures. My results show that contractions in a locally embedded lead sector propagate to upstream suppliers and downstream users through production networks, amplifying employment losses and triggering broader industrial decline (Acemoglu et al., 2012; Barrot & Sauvagnat, 2016; Carvalho, 2014; Gagliardi et al., 2023). These propagation dynamics also help explain slow or incomplete adjustment to large shocks documented in recent work (Autor et al., 2016; Berbée et al., 2025; Dix-Carneiro & Kovak, 2017; Esposito & Abramson, 2021; Franck & Galor, 2021; Fritzsche & Wolf, 2023; Glaeser et al., 2015; Jacobsen & Parker, 2016). These findings highlight the importance of forward-looking transition strategies, especially under planned shifts such as decarbonization, where coordinated policies may mitigate network spillovers (Dechezleprêtre & Sato, 2017; Walker, 2013).

Third, this paper contributes to the literature on regional adjustment to industry shocks by identifying production linkages as a key mechanism shaping the long-term local consequences of industrial expansion and decline. While prior work has documented the persistent effects of trade shocks, plant closures, and resource booms on local labor markets, the channels through which these shocks propagate are still not fully understood (Autor et al., 2016; Black et al., 2005; Feyrer et al., 2017; Greenstone et al., 2010; Hooker & Knetter, 2001). By comparing two closely related industries that differed in their degree of local supply chain integration, I isolate the role of production linkages and demonstrate that they play a decisive role in determining how industry shocks affect local labor markets.

Fourth, it adds to work on the heterogeneity of natural resource extraction impacts by showing that the direction and magnitude of local effects depend on the degree of economic integration

between the extractive sector and the surrounding economy (Allcott & Keniston, 2018; Aragón & Rud, 2013; Black et al., 2005; Lippert, 2014; Pelzl & Poelhekke, 2021). This insight also informs the broader resource-curse debate by showing that variation in local linkage structures helps explain why resource extraction can lead to divergent long-run outcomes (Cust & Poelhekke, 2015; Havranek et al., 2016; Michaels, 2011; Sachs & Warner, 2001; van der Ploeg, 2011).⁹

Structure of the paper. The remainder of the paper is organized as follows. Section 2 provides background on Germany’s coal mining industries, including evidence on differences in local supply chain integration between the hard coal and lignite sectors. Section 3 develops the conceptual framework and derives testable hypotheses, drawing on the local multiplier model of Moretti (2010). Section 4 describes the data, and Section 5 outlines the empirical strategy and identification approach. Section 6 presents the main findings, focusing on the long-run economic impacts of coal mining, the cyclical effects of resource booms and busts, and the mechanisms driving the effects. Section 7 assesses the robustness of the results. Section 8 discusses the implications and concludes.

2 Setting: Hard Coal and Lignite Mining in Germany

Germany provides a compelling setting to examine how variation in production linkages shapes the long-run local effects of resource extraction. For over a century, the country extracted two major coal types: hard coal (anthracite and bituminous) and lignite (brown coal), both produced under a unified institutional and regulatory framework. Although the two industries operated at comparable national scales and during overlapping periods, they differed substantially in their degree of integration into the surrounding economy. Hard coal was closely linked to upstream supplier industries and downstream industrial users, while lignite remained more economically isolated. This contrast allows for a within-country comparison of resource sectors that differ in linkage intensity but are otherwise similar in scale and institutional context. Thereby, it offers a credible setting to isolate the role of production linkages in shaping long-run regional development, during both the expansion and decline of extractive activity.

⁹A related literature links resource booms to higher risks of violence and conflict, offering an additional channel for adverse local effects (Berman et al., 2017; Caselli & Michaels, 2013; Dal Bó & Dal Bó, 2011; Dube & Vargas, 2013).

2.1 Evolution of Coal Mining in Germany

While coal extraction in Germany began in the 18th century, it remained limited in scale until the mid-19th century. From that point onward, the diffusion of coal-based technologies from Britain both stimulated industrial demand and facilitated the expansion of mining activity (Tilly & Kopsidis, 2020).¹⁰ Coal soon became the lead energy source for Germany’s industrial economy.¹¹ By the early 20th century, Germany had become one of Europe’s largest hard coal producers, with output levels comparable to Britain. Germany also emerged as the leading producer of lignite in Europe (Fremdling, 1995). While hard coal production initially dominated, lignite output caught up during the interwar period. From the 1920s onward, hard coal and lignite were produced at roughly similar scales (see Figure A1).¹²

Germany’s mining sector played a central role in the economies of coal-producing regions, both as a major employer and a land-intensive industry. At the beginning of the 20th century, more than 600,000 workers were employed across over 800 mines in the German Empire (Fischer, 2011). In 1907, mining accounted for over 8 percent of total employment across coal-producing counties.¹³ In addition to its labor demands, coal mining required substantial land for extraction and infrastructure, making it both labor- and land-intensive.

By the 1960s, German hard coal production declined strongly. High labor costs, competition from cheaper imported coal, and the growing adoption of alternative energy sources, such as oil, natural gas, and nuclear power, undermined its competitiveness. In particular, the influx of cheap oil from the late 1950s onward, mostly from Northern Africa, triggered a negative labor demand shock in mining (Fritzsche & Wolf, 2023). Between 1960 and 1985, hard coal production fell by nearly 50 percent, roughly 85 percent of mines closed, and employment in the sector declined by more than two-thirds.¹⁴ The last hard coal mine in Germany closed in 2018. Lignite remained

¹⁰The number of steam engines in Prussia increased from 1,445 (29,482 horsepower) in 1849 to 35,431 (958,366 horsepower) in 1878, largely increasing demand for coal and facilitating its extraction (Banken, 1993; Tilly & Kopsidis, 2020).

¹¹Between 1850 and 1900, Germany’s hard coal production increased more than 20-fold, while lignite production rose more than 26-fold (see Figure A1a). Over the same period, employment in hard coal mining rose more than 12-fold, and lignite employment increased over 8-fold (own calculations based on data from Fischer (2011)).

¹²This catch-up was enabled by key technological innovations, most notably Exter’s lignite briquetting press, which improved the energy density and transportability of lignite.

¹³Own calculations based on data from Kaiserliches Statistisches Amt (1910); see Section 4.2 for the definition of coal regions.

¹⁴Own calculations based on data from Fischer (2011).

economically viable until the early 1990s, as technological improvements in surface mining, delayed its decline. By the 1990s, lignite output began to decline as stricter environmental regulations were introduced. Between 1990 and 2023, lignite output declined by more than two-thirds.¹⁵ Although lignite extraction remains active today (as of 2025), it is scheduled to be fully phased out by 2038 as part of Germany’s broader energy transition (KVBG, 2020). This long historical trajectory enables analysis of both the short- and long-run effects of extractive activity and its subsequent decline.

2.2 Institutional Context

In Germany, hard coal and lignite were extracted under similar legal and administrative regimes. In the Prussian territories, including the Ruhr, Rhineland, and Lower Lusatia, the 1851 Co-ownership Act (*Miteigentümerge-setz*) transferred mine management from the state to private actors, incentivizing investment, technological advancement, and large-scale coal extraction (Tilly & Kopsidis, 2020). This was followed by the 1865 General Mining Law (*Allgemeines Berggesetz*), which established uniform rules for mining concessions that applied equally to hard coal and lignite (ABG, 1865). Most other German states (e.g., Baden, Bavaria, Hesse, Württemberg) adopted similar legislation modeled on the Prussian framework (Deutscher Bundestag, 2024).¹⁶ This institutional alignment across coal types suggests that differences in long-run local outcomes between hard coal and lignite regions are unlikely to stem from variation in regulation or governance. Instead, they plausibly reflect differences in production linkages.

German coal extraction did not generate direct fiscal windfalls for municipalities through royalties. While local governments collected business tax revenue from local firms (*Gewerbesteuer*)¹⁷ mining royalties accrued to the federal states (ABG, 1865; BBergG, 1980). This institutional arrangement limits the scope for direct fiscal linkages at the municipal level. Moreover, taxation rules did not differ between hard coal and lignite mining, suggesting that differential fiscal treatment is unlikely to explain observed variation in long-run outcomes across mining regions (Gunton, 2003; Watkins, 1963).

¹⁵Own calculations based on data from Statista Research Department (2025a).

¹⁶A nationwide legal consolidation only occurred with the Federal Mining Act (*Bundesberggesetz*) in 1982 (BBergG, 1980).

¹⁷The *Gewerbesteuer* was introduced in 1891 (Preußisches Gewerbesteuergesetz, 1891).

2.3 Geographic Distribution of German Coal Resources

Coal extraction in Germany was geographically concentrated in several distinct regions, with the two coal types—hard coal and lignite—mined in largely non-overlapping parts of the country. Major hard coal deposits were located in central and western Germany, particularly in the Ruhr region (e.g., the districts of Arnsberg and Düsseldorf), the Saar area (e.g., Trier), and in Upper Silesia (e.g. Oppeln, now part of modern Poland). In contrast, lignite was mined mainly in the Rhenish mining district (e.g., Köln), the Lusatian district (e.g., Frankfurt-Oder), and the Central German mining district (e.g., Merseburg) (Kunz, 2008). Figure 3 illustrates the distribution of coal fields in Germany, which closely corresponds to the historical location of coal mines (see also Section 4.2). The clear geographic separation between hard coal and lignite deposits, combined with their broad dispersion across Germany, provides the variation needed to credibly compare local economic outcomes across regions exposed to hard coal, lignite, or no coal mining, while holding institutions constant.

2.4 Differences in Production Linkages Between Hard Coal and Lignite

Hard coal and lignite mining differed markedly in their potential to generate local economic spillovers through input–output linkages. I distinguish between two types: upstream linkages, reflecting the mining sector’s demand for local intermediate inputs, and downstream linkages, reflecting its role as a supplier to other local industries.

2.4.1 Upstream Linkages: Coal Mining’s Dependence on Local Inputs

Hard coal mining relies more heavily on upstream industries and services than lignite mining, reflecting its more complex extraction process. Unlike lignite, which is typically found near the surface and extracted via open-pit methods, hard coal lies deep underground. Extracting it requires shafts, tunnels, drainage systems, and specialized equipment.¹⁸ This more demanding extraction process creates greater demand for machinery, metal components, industrial materials, and related services (Hovis & Mouat, 1996; Pierenkemper, 1992).

Figure 1 illustrates this stark difference in input intensity. In 1936, intermediate goods accounted

¹⁸See Figure A4 for photographs illustrating the extraction methods of lignite and hard coal mining.

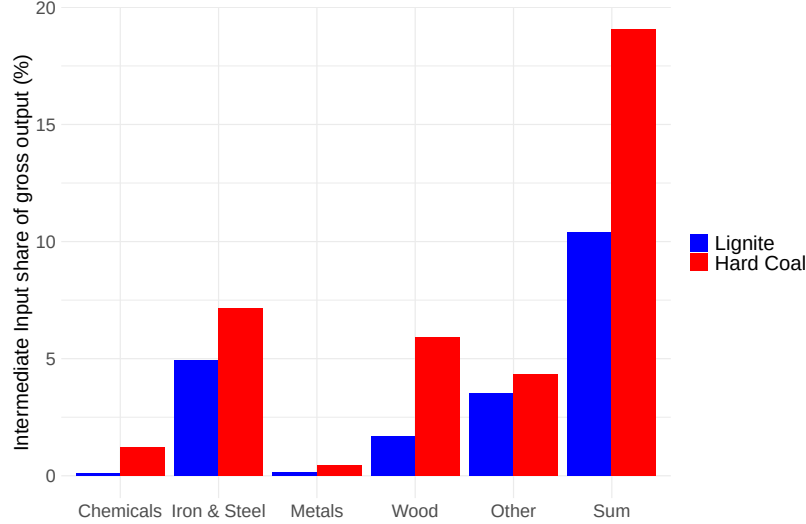


Figure 1: Hard Coal and Lignite Mining Intermediate Inputs in Germany (1936)

Notes: The figure plots the value of inputs purchased from various sectors by the lignite (blue) and hard coal (red) mining industries in Germany in 1936, expressed as a share of each sector’s total output. Own calculations based on data from the Statistisches Reichsamt (1936a).

for 19.1% of total production value in hard coal mining, compared to just 10.4% in lignite. Hard coal also required a higher share of specialized materials, such as mine timber and explosives, which made up 6.3% of hard coal output versus only 1.5% for lignite (Statistisches Reichsamt (1936a), own calculations).

Where economic and institutional conditions permit, mining firms often source inputs locally to reduce transport costs, improve reliability, and allow for customized supply (Ellison & Glaeser, 1997; Moretti, 2011; Morris et al., 2012). Consequently, the greater input intensity of hard coal extraction results in stronger upstream linkages to local suppliers compared to lignite.

2.4.2 Downstream Linkages: Local Use of Coal

Hard coal supports a broader range of industrial applications than lignite, particularly in energy-intensive sectors, resulting in stronger downstream linkages. By contrast, lignite’s lower energy density, higher moisture content, and lower carbon concentration limits its industrial use.¹⁹

The differences in energy content and usability between hard coal and lignite translates into

¹⁹Lignite’s chemical composition also prevents its use in coke production, a crucial input for blast furnaces and heavy industry.

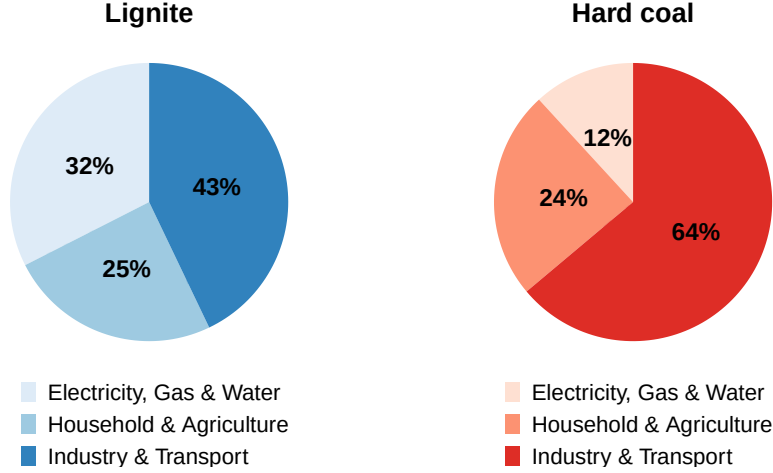


Figure 2: Hard Coal and Lignite Use in Germany (1934)

Notes: The figure shows the sectoral shares of lignite (left) and hard coal (right) use in Germany in 1934. Own calculations based on data from the Statistisches Reichsamt (1934).

distinct patterns of industrial coal use, as shown in Figure 2. Lignite consumption was more heavily concentrated in electricity generation, with relatively smaller use in industrial production compared to hard coal. Hard coal, by contrast, was primarily used in industrial processes, particularly in heavy manufacturing, metal processing, and transport.

Processing coal near extraction sites also offered important economic advantages. Coal is bulky, degrades quickly, and is expensive to transport and store. These constraints encourage spatial co-location between coal extraction and industrial production, particularly in hard coal regions (Ellison & Glaeser, 1997). As a result, downstream spillovers through industrial demand are likely far more pronounced in hard coal regions than in lignite regions.

Taken together, the stark difference in linkage intensity between the hard coal and lignite industries, despite similar institutional contexts, provides a rare opportunity to isolate how input–output structures shape the long-run regional effects of resource extraction.

3 Conceptual Framework: Sectoral Spillovers from Coal Mining

To study how coal mining affects employment across sectors depending on linkage intensity, I develop a simple general equilibrium framework adapted from Moretti (2011). I model each county as a competitive economy that produces tradable goods (mining, manufacturing, agriculture) sold

on national markets and non-tradable goods (services) whose prices are set locally.²⁰ Labor is homogeneous and mobile across sectors within a county, implying that wages equalize locally across industries. The local labor supply curve is upward sloping: workers' utility depends on local wages net of living costs, along with idiosyncratic location preferences. Housing supply is also upward sloping, reflecting geographic constraints and land-use regulation. Amenities are assumed to be constant across space. This framework allows me to derive testable predictions about how extractive sectors with varying linkage intensities shape local labor markets.

Spillovers to the Tradable Sector. An increase in mining employment raises local wages, which in turn increases production costs for all local firms. Since tradable goods are sold in national or international markets, local producers cannot pass these higher costs on to consumers (they are price-takers), potentially leading to crowding-out effects. However, if mining generates upstream or downstream linkages with other local industries, it may stimulate activity in related sectors. The net impact depends on the strength of these production linkages and the presence of agglomeration economies, such as productivity gains from industry clustering.

In the case of hard coal, both upstream and downstream linkages are strong, as outlined in Section 2.4. Mining depends on industrial inputs and simultaneously supplies fuel to a broad range of industries. These linkages are likely to generate positive spillovers to nearby manufacturing firms, which may expand employment in response to increased input demand from mining or improved access to coal as an input. By contrast, agriculture, lacking direct connections to mining, primarily faces cost-side pressures: rising wages and land prices reduce profitability, potentially leading to employment declines.

Hypothesis H1: *Manufacturing employment increases in hard coal regions, particularly in sectors with strong input–output linkages.*

Hypothesis H2: *Agricultural employment declines in hard coal regions due to rising input costs and land competition.*

In lignite regions, production linkages are relatively weak. Lignite is primarily used for local heating and electricity generation, and its extraction process relies on fewer intermediate inputs.

²⁰This distinction follows Moretti (2011), based on the geographic scope of product markets.

As a result, manufacturing firms benefit little from proximity to mining activity but may still face upward wage pressure, potentially reducing employment. The effects on agriculture are more ambiguous. On the one hand, rising wages may increase production costs; on the other hand, if mining displaces other sectors and weakens overall labor demand, wage pressures may ease over time, potentially stabilizing agricultural employment.

Hypothesis L1: *Manufacturing employment in lignite regions declines due to weak linkages and rising wages.*

Hypothesis L2: *Agricultural employment in lignite regions remains stable or increase if wage pressures are offset by reduced demand in other sectors.*

Spillovers to the Non-Tradable Sector. Higher mining wages and employment raise local incomes, which in turn increase demand for non-tradable goods and services. The magnitude of this demand linkage effect depends on the labor intensity of services and the share of income spent locally. In addition, mining operations may generate direct demand for services such as logistics, maintenance, and catering.²¹

In hard coal regions, both income-driven and mining-related service demand are expected to raise employment in the non-tradable sector. In lignite regions, by contrast, weaker income gains and limited service linkages may result in stagnant or declining service employment, particularly if population falls due to out-migration or crowding-out of other sectors.

Hypothesis H3: *Service employment increases in hard coal regions due to higher income and mining-related demand.*

Hypothesis L3: *Service employment stagnates or falls in lignite regions due to limited income growth and weak service demand from mining.*

Total Employment Effects. In regions with strong input–output linkages, mining acts as a catalyst for broader economic expansion. Hard coal regions should experience net employment growth across sectors, despite agricultural losses. In lignite regions, weak spillovers and higher costs may limit overall gains or lead to net employment losses.

²¹Moretti (2010) estimates that each additional manufacturing job generates approximately 1.6 jobs in local services.

Hypothesis H4: *Total employment increases in hard coal regions due to strong local multipliers.*

Hypothesis L4: *Total employment effects in lignite regions are modest or negative due to weak spillovers and crowding-out effects.*

Effects of Coal Phase-Outs. The framework also applies to gradual declines in mining activity, such as those associated with long-term coal phase-outs. These contractions reverse the mechanisms described above.

In hard coal regions, strong upstream and downstream linkages imply that declining mining employment reduces demand in manufacturing and services, triggering broader employment losses. In lignite regions, by contrast, impacts are more concentrated within mining itself. Still, falling incomes and population may reduce demand for local services, leading to additional employment losses in the non-tradable sector.

Hypothesis H5: *A decline in hard coal mining reduces employment in manufacturing and services due to strong production and demand linkages. Effects on agriculture are ambiguous and may depend on how wage and land cost pressures adjust.*

Hypothesis L5: *In lignite regions, employment losses are concentrated in services, while manufacturing and agriculture are less affected.*

I test these predictions using two complementary empirical strategies. First, I employ an event-study framework to examine the long-run effects of mining and the coal phase-out. Second, I implement a shift-share design that interacts global coal price shocks with county-level exposure to coal endowments, capturing the dynamic effects of resource booms. Together, these strategies allow me to isolate the causal impact of coal extraction and distinguish between short-run boom-bust cycles and persistent long-term effects.

4 Main Data

This section describes the construction of the main dataset used in the analysis. The unit of observation is the county. To ensure spatial consistency over time, I harmonize administrative boundaries to match those in place in 1900 (see Appendix A.3).

4.1 Outcome Variables: County-Level Sectoral Employment

To analyze the local effects of coal mining, I compile a novel county-level employment dataset for Germany spanning 1849 to 2024. The dataset captures the rise and expansion of the hard coal and lignite industries, as well as the onset of coal phase-outs beginning in the 1960s. To construct a consistent panel, I harmonize sectoral employment classifications across 28 censuses and align administrative boundaries from Prussia to modern-day Germany (see Section A.3). To my knowledge, this is the first dataset to track local employment in Germany at the county level across such a broad temporal and industrial scope.²² This long-run panel allows me to trace the economic effects of coal mining during both the extraction period and the subsequent decline.

Disaggregated Manufacturing Employment and Production Linkages. To examine heterogeneity in the effects of coal mining across manufacturing industries, I construct measures of upstream and downstream linkages between each industry and the mining sector. These linkage intensities are derived from the 1947 U.S. Input-Output (I-O) table compiled by the U.S. Bureau of Economic Analysis (2019), which provides detailed information on inter-industry transaction flows.²³

For each manufacturing industry, I calculate first- and second-degree upstream and downstream linkage shares. The first-degree downstream linkage is defined as the share of an industry’s total inputs sourced directly from the mining sector:²⁴

$$\text{FirstDegreeDownstream}_i = \frac{\text{InputFromMining}_i}{\text{TotalInput}_i}$$

The first-degree upstream linkage is defined as the share of an industry’s output sold directly to the mining sector:

$$\text{FirstDegreeUpstream}_i = \frac{\text{OutputToMining}_i}{\text{TotalOutput}_i}$$

Second-degree linkages capture indirect relationships via intermediate industries and are calculated analogously through two-step I-O matrix multiplications. An industry is classified as

²²Descriptive statistics are reported in Table A2. Section A.4 describes the underlying sources.

²³I rely on the U.S. I-O table to avoid potential endogeneity concerns associated with using historical German tables, which may themselves have been shaped by coal mining activity. In robustness checks, I confirm that the results are not sensitive to the choice of I-O table by replicating the analysis using a historical German benchmark.

²⁴The mining sector itself is excluded from linkage calculations.

upstream-linked if its total upstream linkage (direct plus indirect) exceeds the sector median. To prevent electricity-related sectors from dominating the downstream classification, I follow Allcott and Keniston (2018) and use only the direct input share.²⁵ Industries are considered downstream-linked if their direct input share from mining exceeds the median. An industry is defined as linked to mining, if it is classified as either upstream- or downstream-linked; otherwise, it is considered non-linked.

Appendix Table A4 reports the linkage shares and binary indicators for each manufacturing industry. The strongest upstream linkages are found in *Machinery* and *Wood products*, both of which supply key inputs to mining operations. Industries with the highest downstream linkages are *Nonmetallic mineral products* and *Primary metals*, reflecting their reliance on coal as an energy input. Finally, I map census occupations for each available year to the I-O sectors, allowing disaggregated analysis of sector-specific employment dynamics.²⁶

Sectoral Diversification of Manufacturing Employment. To assess whether coal mining influenced the structure of local economies, I examine its effect on industrial diversification across regions. I measure diversification using a Herfindahl–Hirschman Index (HHI) of manufacturing employment, calculated at the county-year level for each census year in which disaggregated occupational data are available. The index is defined as the sum of squared employment shares across manufacturing industries:

$$\text{HHI}_{ct} = \sum_{i \in I} \left(\frac{\text{Employment}_{ict}}{\text{TotalEmployment}_{ct}} \right)^2,$$

where $i \in I$ indexes manufacturing industries observed in county c and year t . Higher index values reflect greater concentration, while lower values indicate a more diversified manufacturing structure. This measure allows me to examine whether coal extraction led to increased industrial specialization or diversification in coal-endowed regions, and whether these structural patterns shaped the effects of the subsequent coal phase-out.

²⁵As shown in Section 7, results are similar when using either direct or indirect input shares for both upstream and downstream linkages.

²⁶While I observe total manufacturing employment data for 28 census years between 1849 and 2024, disaggregated industry-level data (at the 2- or 3-digit level) are only available for the years 1849, 1882, 1895, 1907, 1939, and from 2007 to 2024.

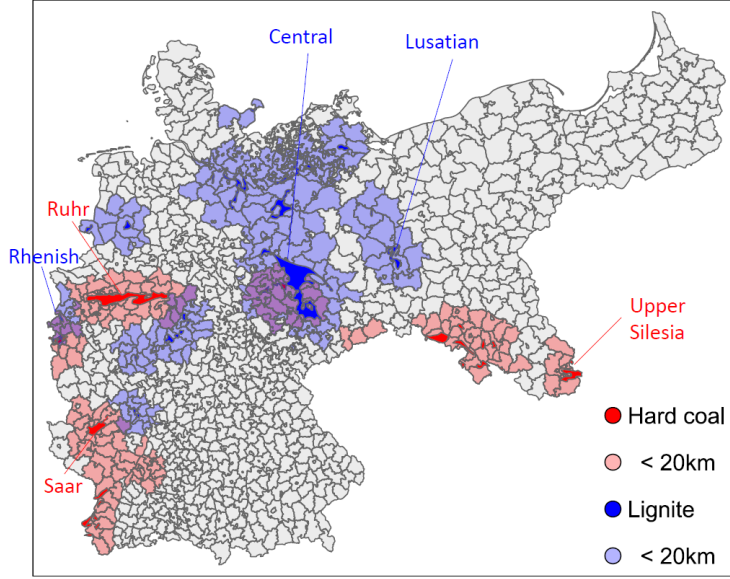


Figure 3: Geographic Locations of Hard Coal and Lignite Deposits in Germany

Notes: The map displays hard coal (red) and lignite (blue) rock strata within the borders of Germany as of 1900. Counties classified as hard coal or lignite regions in the analysis, i.e. those located within 20 kilometer (km) of the respective deposits, are highlighted accordingly. Major coal-producing areas (e.g., Ruhr, Saar, Upper Silesia) and lignite basins (e.g., Rhenish, Central German, Lusatian) are labeled. Data source: Asch and Bellenberg (2005).

4.2 Treatment Variables: Hard Coal and Lignite Endowments

To measure exposure to coal mining, I use geological data on the location of hard coal and lignite rock strata across Germany. Rather than relying on historical mining activity, which may be endogenous to prior economic development, I follow A.-K. Becker and Hornung (2025), De Pleijt et al. (2020), Esposito and Abramson (2021), and Fernihough and O’Rourke (2021) in using geological resource endowments, based on data from Asch and Bellenberg (2005), as a plausibly exogenous proxy for mining potential.

I classify strata from the Carboniferous period (ca. 320–295 million years ago) as hard coal-bearing, since nearly all German hard coal was formed during this era. Lignite, by contrast, is typically found in younger formations dating from the Miocene to Pliocene periods (ca. 60–5 million years ago). Because these younger formations are geologically more heterogeneous, I restrict lignite classification to areas where lignite is among the two most prevalent rock types. I then compute the distance from each county’s geographic centroid to the nearest hard coal or lignite-bearing strata. As shown in Figure A2a and Figure A2b, the distances from county centroids to hard coal

and lignite fields are highly correlated with the distances from county centroids to carboniferous and lignite rock strata, respectively. I classify a county as a hard coal (or lignite) region if its centroid lies within 20 kilometers of the corresponding strata. Alternative thresholds are tested in robustness checks (see Section 7).

Table A1 presents summary statistics. On average, counties are located 132 kilometers from the nearest hard coal deposit and 110 kilometers from the nearest lignite deposit. Based on the 20 kilometers cutoff, 16% of counties are classified as hard coal regions and 18% as lignite regions. About 30% of counties are located near either type of deposit, with roughly 3% lying near both.

4.3 Geographic Controls

I account for several geographic factors that may confound the relationship between coal endowments and local labor market outcomes. First, to account for differences in market access, I control for the distance to the nearest navigable river and the coast, both of which may influence employment patterns and may be correlated with coalfield locations. Second, I include climatic and agricultural suitability measures, average temperature, precipitation, and a caloric suitability index from Galor and Özak (2016), to capture variation in agricultural productivity that may affect sectoral employment. Third, I control for terrain slope to account for the historical use of water mills, the main alternative to steam engines prior to widespread coal adoption (Gutberlet, 2014). Steeper terrain was more suitable for water-powered mills, which facilitated early proto-industrial development. As a result, these regions may have accumulated more physical capital and been better positioned to adopt new technologies, with potential long-run effects on employment patterns (Ashraf et al., 2025).²⁷

²⁷Table A1 provides descriptive statistics.

5 Empirical strategy

5.1 Event-study: Long-Run Effects of Resource Extraction and Its Phase-Out

To assess the long-run effects of hard coal and lignite mining on local economies and to evaluate the effects of the coal phase-out, I estimate the following event-study specification:

$$Y_{ct} = \sum_{j=t_1}^T \beta_j (HardCoal_c \times I_t^j) + \sum_{j=t_1}^T \delta_j (Lignite_c \times I_t^j) + \sum_{j=t_1}^T (X_c' I_t^j) \kappa_j + \alpha_c + \gamma_t + \varepsilon_{ct}. \quad (1)$$

Here, Y_{ct} is the outcome of interest in county c and year t .²⁸ The indicators $HardCoal_c$ and $Lignite_c$ equal 1 if the county centroid lies within 20 kilometers of a hard coal or lignite deposit, respectively. These indicators are interacted with year dummies I_t^j to estimate time-varying treatment effects. The coefficients β_j and δ_j represent the differential outcomes for hard coal and lignite counties, respectively, compared to non-coal counties in year j , relative to the baseline year. Due to changes in territorial boundaries, I estimate the model separately for (i) the period of coal industry expansion and (ii) the era of rapid coal decline, using distinct time windows and baseline years for each subsample.

The vector X_c includes time-invariant geographic controls (temperature, rainfall, slope, crop suitability, distance to the nearest river and coast) interacted with year fixed effects to flexibly account for time-varying confounders. County fixed effects α_c absorb all time-invariant unobserved heterogeneity, while year fixed effects γ_t control for common shocks affecting all counties. Standard errors are clustered at the county level.

Identification relies on the standard parallel trends assumption. Let treatment be the positive labor demand shock associated with the rapid expansion of industrial-scale coal mining in Germany beginning in the mid-19th century. In the absence of this event, treated counties (with coal deposits) and control counties (without deposits) would have followed similar trends in the outcome, conditional on controls and fixed effects. This assumption is plausible, as coal deposits were determined by natural processes long before the onset of modern economic activity. I further assess the validity of this assumption by examining pre-trends in an alternative outcome (wages) in

²⁸ t is observed in selected years between 1849 and 2024: 1849, 1882, 1895, 1907, 1939, 1950, 1961, 1970, 1987, and annually from 2007 to 2024.

Section 7. For the decline period, treatment captures the negative labor demand shock associated with the contraction of the coal sector, primarily driven by the influx of cheap oil starting in the late 1950s. In this case as well, identification rests on the assumption that, absent the decline, treated and comparison counties would have exhibited parallel trends. I assess this assumption by reporting pre-trends in the main outcomes.

5.2 Reduced form Shift-Share Approach: Effects of Coal Booms

To complement the event-study analysis, I estimate a reduced-form “shift-share” design that exploits plausibly exogenous variation in exposure to global coal market fluctuations. Exposure is measured by proximity to geological coal deposits, while global coal prices provide the time-varying shocks.

$$Y_{ct} = \beta_H \text{HardCoal}_c \times \text{WorldPrice}_t + \beta_L \text{Lignite}_c \times \text{WorldPrice}_t + \sum_{j=t_1}^T X'_c I_t^j \kappa_j + \alpha_c + \gamma_t + \epsilon_{ct}. \quad (2)$$

Here, HardCoal_c and Lignite_c are binary indicators equal to 1 if county c lies within 20 kilometers of a hard coal or lignite deposit, respectively. The variable WorldPrice_t is the (log) annual average world coal price, deflated to constant prices and standardized to mean zero and unit standard deviation (sd).²⁹ It captures the contemporaneous³⁰ price shock in year t .³¹ Coefficients β_H and β_L capture the differential effect of a one sd increase in global prices on outcomes in hard coal and lignite regions, respectively, relative to non-endowed counties. All specifications include county fixed effects (α_c), year fixed effects (γ_t), and a vector of geographic controls interacted with year fixed effects ($X'_c I_t^j$).

This design complements the event-study framework by leveraging a distinct source of identifying variation. While the event study relies on parallel trends, identification here requires that,

²⁹Separate world price series for hard coal and lignite are not available, but both fuels are expected to co-move with global energy prices. Although lignite was little traded internationally, German hard coal and lignite prices exhibit strong year-to-year co-movement (correlation of annual changes $\rho = 0.87$, $p < 0.001$ for the period 1882–1939), suggesting exposure to common cyclical shocks. As a robustness check, I also use national-level series that allow for coal-type-specific shocks (see Section 7).

³⁰To account for potential delayed responses, I use a five-year moving average of the price shock in a robustness check (see Section 7).

³¹Price data are available for all years from 1882 to 2024.

conditional on controls, there is no unobserved factor that is both correlated with a county’s geological coal endowment and systematically co-moves with global coal prices. The exposure measure is plausibly exogenous because the location of coal deposits is time-invariant and determined by ancient geological processes unrelated to modern economic activity (Goldsmith-Pinkham et al., 2020). The shift component, movements in the world coal price, is likewise plausibly exogenous to local German conditions (Adao et al., 2019; Borusyak et al., 2022). To further mitigate concerns, I include county and year fixed effects, a rich set of geographic controls interacted with year dummies, and robustness checks using placebo shocks and alternative shift variables to support the identifying assumption (see Section 7).

The shift–share design also offers several practical advantages. By leveraging exogenous variation from global coal booms and busts, I begin the analysis in 1882. This enables inclusion of all counties within the historical borders of the German Empire, whereas the event-study analysis is restricted to Prussian counties. Additionally, price shocks provide more precise timing of economic cycles than the gradual and continuous treatment used in the event-study. While local labor market responses may not be immediate, due to adjustment frictions, the shift–share design helps distinguish short-run boom–bust dynamics from longer-run structural effects of resource extraction and reinforces identification through an alternative source of variation.

6 Main Results

6.1 Long-Run Effects of Resource Extraction: Event-study Results

I begin by examining the long-run labor market effects of the positive labor demand shock generated by the onset of large-scale coal extraction. The year 1849 serves as the pre-treatment baseline, as coal production was limited to small-scale operations prior to this point. Industrial-scale mining expanded rapidly in the following decades, driven by advances in extraction technologies and rising industrial demand from the adoption of coal-using technologies. I therefore expect potential spillover effects to emerge by the first post-baseline year, 1882 (see Figure A1).³² The event-study estimates thus capture employment changes after the onset of large-scale coal extraction.

³²The analysis is restricted to counties that belonged to Prussia in 1849, as occupational census data for that year are only available for this subset. I end the analysis in 1939 to avoid confounding effects from the major border changes that occurred after World War II.

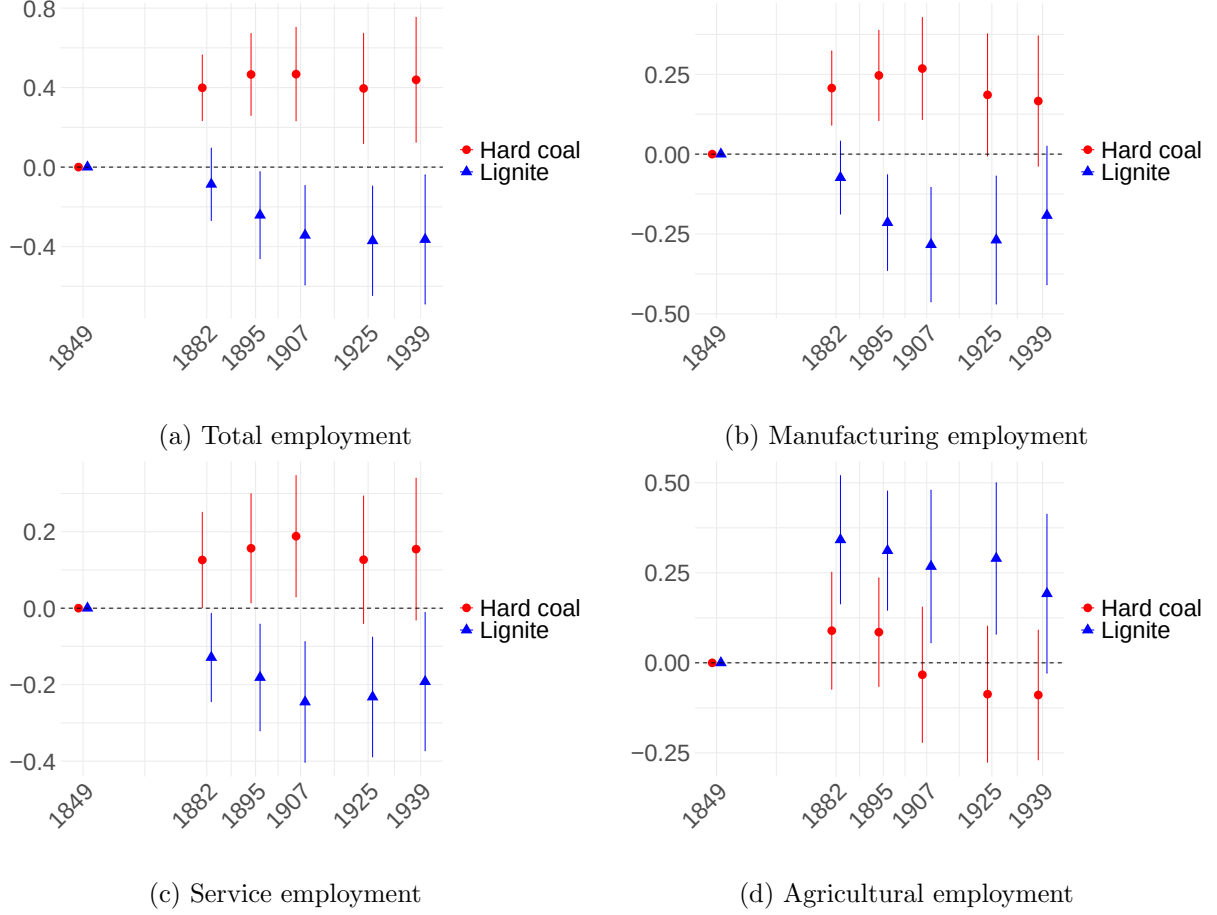


Figure 4: Event-Study: Effect of Resource Endowment on Sectoral Employment (1849–1939)

Note: This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment, (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite endowment indicators are interacted with year fixed effects; the omitted year is 1849. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties. Corresponding estimates for each plot are presented in Table A6 in the Appendix.

Figure 4 shows that the local effects of coal extraction varied sharply by coal type, consistent with the role of production linkages outlined in Section 3. Total employment³³ increased by 0.4 to 0.5 (log) sd between 1849 and 1882 in hard coal regions compared to non-coal regions after 1849 (Panel 4a).³⁴ Employment remained persistently elevated through multiple boom-and-bust

³³Total employment is defined as the sum of agricultural, manufacturing, and service sector employment. Mining employment is excluded, as data for this sector are not consistently reported across all benchmark years and unavailable for the baseline year (1849).

³⁴Although the 1849 data point alone does not permit a formal pre-trend test, the discrete employment shift between 1849 and 1882, followed by a stable divergence through 1939, is consistent with the interpretation that these differences were caused by the onset of industrial-scale coal mining rather than pre-existing trends. I further assess

periods until 1939.³⁵ In contrast, lignite regions saw a sustained decline between 1882 and 1939 below baseline, consistent with limited mining spillovers and potential crowding-out of other sectors. These patterns align with **Hypothesis H4**, which anticipates broad employment gains in hard coal regions following large-scale mining expansion, driven by strong production linkages and local multipliers. They are also consistent with **Hypothesis L4**, which predicts limited or negative effects of lignite extraction due to weak intersectoral connections. In manufacturing (Panel 4b), employment in hard coal regions rose by 0.2 to 0.3 (log) sd, in line with strong upstream and downstream linkages (**Hypothesis H1**). Lignite regions saw declines of similar magnitude, consistent with weak production linkages and crowding-out effects (**Hypothesis L1**). A similar pattern emerges in the service sector (Panel 4c): employment in hard coal regions rose, likely reflecting demand spillovers from rising population, income, and mining-related service needs (**Hypothesis H3**). By contrast, service employment in lignite regions declined, likely due to a shrinking workforce, weaker local demand, and limited demand-side linkages from mining activity (**Hypothesis L3**). In agriculture (Panel 4d), hard coal regions show no significant change, consistent with offsetting forces from wage pressures (**Hypothesis H2**). In contrast, agricultural employment in lignite regions increases modestly, likely reflecting reallocation into lower-productivity sectors as manufacturing and services contracted (**Hypothesis L2**).

To assess the magnitude of these effects more transparently, Table A7 reports estimates from a difference-in-differences specification that replaces the event-time indicators in Equation 1 with a post-1849 indicator. Relative to 1849, hard coal regions experienced increases of 0.42 (log) sd in total employment (approximately 25% in levels), 0.21 (log) sd in manufacturing (19%), and 0.14 (log) sd in services (16%). By contrast, lignite regions saw a decline of 0.22 (log) sd in manufacturing employment, equivalent to roughly 20% in levels.

These findings highlight how the strength of local production linkages shaped the long-run regional effects of extraction. In hard coal areas, strong integration into industrial supply chains supported broad-based growth, that persisted across multiple boom and bust cycles. In lignite regions, by contrast, weak linkages limited spillovers and led to structural decline.

robustness in Section 7.

³⁵Despite an overall rise in hard coal output between 1882 and 1939, the employment gap between hard coal and non-coal regions does not widen further. This pattern is likely explained by capital deepening and productivity gains in both mining and coal-linked industries, which reduced labor demand per unit of output.

6.2 Effects of Resource Booms: Estimates of the Reduced-Form Shift-Share Approach

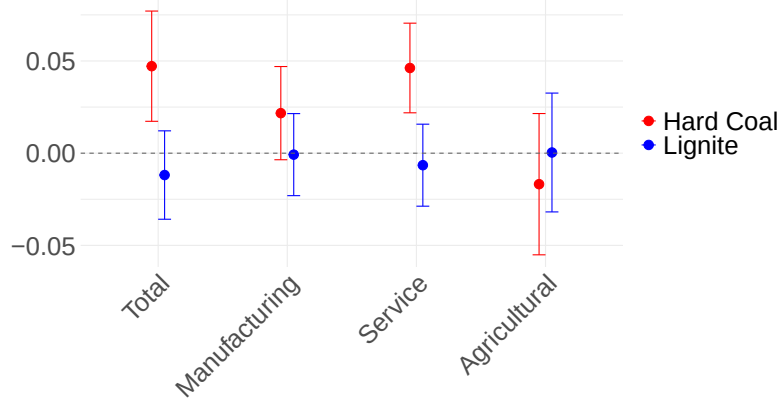


Figure 5: Reduced-form Shift-Share Approach: Effect of Resource Booms on Sectoral Employment (1882–1939)

Note: This figure plots coefficients and 95% confidence intervals from reduced-form shift-share approach based on Equation 2. Each estimate captures the differential effect of changes in global coal prices on counties with hard coal (red) and lignite (blue) endowments, relative to non-endowed counties. The dependent variables are county-level employment in total, manufacturing, services, and agriculture. All outcomes are log-transformed and standardized to have a mean of zero and a sd of one. Standard errors are clustered at the county level. The analysis covers the period 1882–1939 within the borders of the German Empire.

I now turn to the estimates from the reduced-form shift-share approach from Equation 2, which complement the event-study results by exploiting time-series variation in global coal prices interacted with local geological coal endowments.³⁶ This approach allows me to estimate how county-level labor markets responded to coal booms, holding fixed local characteristics and national trends.³⁷

Figure 5 presents the estimated effects of global coal price shocks on sectoral employment in counties with hard coal (red) and lignite (blue) deposits, relative to non-endowed counties. An increase in global coal prices leads to a significant increase in total employment in hard coal regions relative to non-coal regions, suggesting that booms boosted local labor demand. These gains are concentrated in the service sector, consistent with the hypothesis that mining activity generates local demand spillovers. Manufacturing employment also responds marginally significantly in the

³⁶Figure A3 in the Appendix plots annual global coal prices, illustrating boom and bust periods. Descriptive statistics are reported in Table A3.

³⁷In the main specification, I use the exact global coal price in the year employment is observed. The results are robust to using a 5-year moving average of coal prices instead (see Figure A17d in the Appendix).

short run, but the effect is smaller than in services. This likely reflects adjustment frictions and the delayed responsiveness of capital-intensive tradable sectors. Agricultural employment declines slightly, consistent with a gradual structural shift away from farming, although the effect is not statistically significant. The magnitudes of the coefficient estimates are relatively modest. This is consistent with the nature of the shift–share design, which captures short-run marginal effects of price fluctuations, rather than the cumulative effects of large-scale structural transformation. In lignite regions, by contrast, price shocks are not associated with significant changes in employment in any sector. This supports the interpretation that lignite was weakly integrated into local production networks and generated limited upstream, downstream, or demand-side spillovers, consistent with the enclave nature of the industry. The absence of short-run negative effects likely reflects a temporary boost from mining-related demand, which may offset crowding-out pressures in other sectors.

Taken together, the results of the shift–share approach provide further evidence that coal booms in hard coal regions increased local labor demand, while failing to generate similar effects in lignite regions. The smaller coefficient estimates reflect the design’s focus on short-run labor market responses to global price shocks, in contrast to the event study, which captures the cumulative impact of resource extraction. Moreover, the results indicate that non-tradable sectors react more immediately to resource booms, while employment in tradable sectors adjusts more gradually, consistent with slower reallocation dynamics and higher adjustment costs. This alternative identification strategy reinforces the central role of production linkages in shaping the local effects of resource extraction.

6.3 Effects of Resource Phase-Out

So far, I have shown that hard coal mining generated significant economic benefits for producer regions during the extraction period. But what happens when extraction rapidly declines? Do former mining regions retain an industrial advantage through agglomeration forces, or do these benefits erode once extraction ceases?

To examine the local effects of rapid mining decline, I re-estimate the event-study specification from Equation 1, setting 1961 as the baseline year. Around this time, hard coal counties experienced a sustained negative labor demand shock in mining, driven by increased competition from cheap

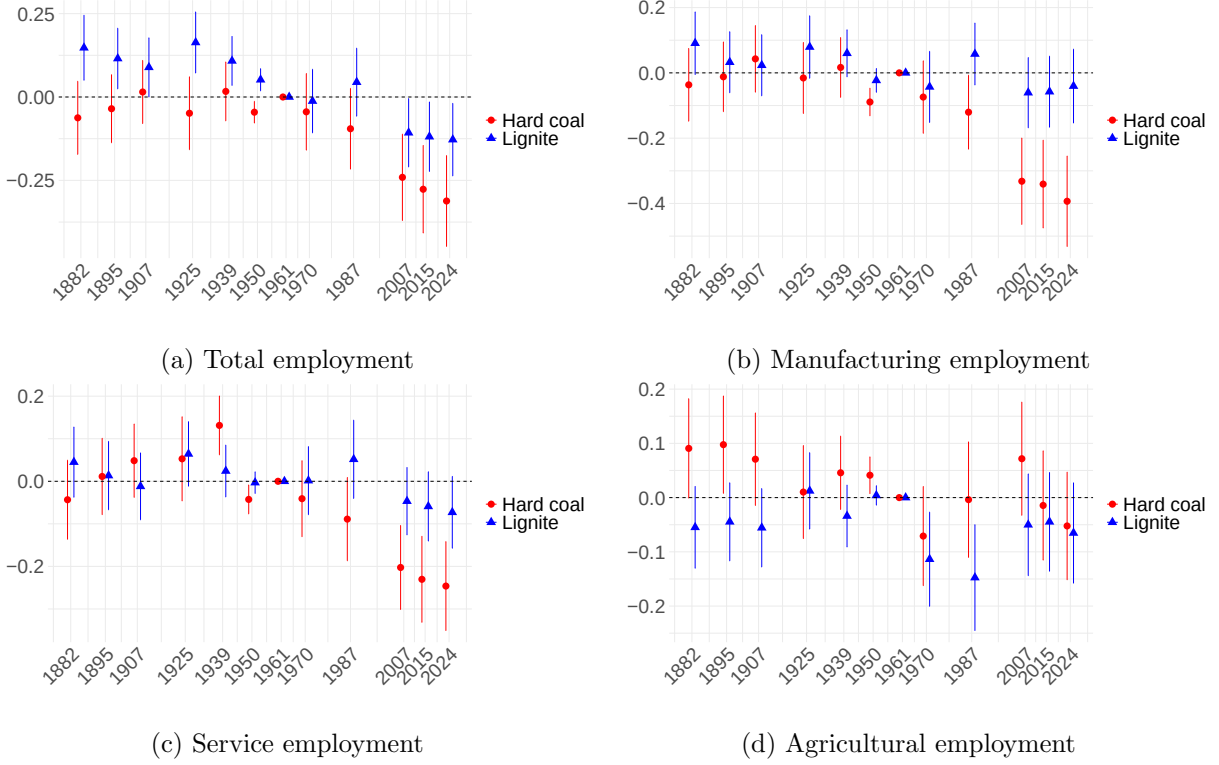


Figure 6: Event-Study: Effect of Coal Phase-Out on Sectoral Employment (1882–2024)

Note: This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment, (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite indicators are interacted with year fixed effects; the omitted year is 1961. Standard errors are clustered at the county level. The analysis is restricted to West German counties.

oil imports and the resulting contraction of domestic production. The lignite phase-out did not begin until the early 1990s and remains ongoing as of 2025 (see Section 2.1). Therefore, the coefficients for lignite regions should be interpreted as employment dynamics over time relative to the 1961 baseline, rather than as responses to a contemporaneous phase-out. Separate regressions for hard coal and lignite, using reference periods corresponding to the respective onset of decline, are included in Figure A7 in the Appendix.³⁸ For the period before 2007, I use manufacturing employment data for all available years from 1882 onward: 1882, 1895, 1907, 1925, 1939, 1950, 1961, 1970, and 1987. From 2007 onward, annual data are available. To maintain comparability

³⁸Specifically, I estimate the event-study separately for hard coal and lignite regions relative to non-hard coal and non-lignite regions, and also when excluding the other coal type from the comparison group. Figure A7 in the Appendix confirms the main findings. Manufacturing employment declines after 1961 in hard coal regions, while lignite regions do not experience a large decline after 1987.

with earlier periods and avoid overrepresenting recent years, I select three reference points: 2007 (first year), 2015 (midpoint), and 2024 (latest year), thereby preserving roughly consistent spacing over time.³⁹

The results indicate that the economic advantages associated with hard coal production did not persist. Figure 6 plots coefficients for total, manufacturing, service, and agricultural employment, comparing hard coal and lignite regions to non-coal regions in West Germany over time. In manufacturing (Panel 6b), I find a clear and persistent decline in employment in hard coal regions relative to non-coal regions after 1961. The negative trend suggests that industrial activity in these regions eroded alongside the contraction of coal mining. The observed pre-trend largely supports the identifying assumption of parallel trends, with no evidence of differential employment trajectories prior to 1961. By contrast, I find no significant employment effects in lignite regions. The estimates remain close to zero and are statistically insignificant over the entire sample period. While lignite extraction is not fully phased out by 2025, it declined substantially over the past three decades (see Section 2.1). The absence of meaningful employment losses in manufacturing, even in later decades, reinforces the interpretation that lignite mining was only weakly integrated into the local production structure, resulting in more muted labor market impacts as production declined.

The other panels of Figure 6 show similar sectoral patterns. In the service sector (Panel 6c), hard coal regions experience a relative decline in employment after 1961 relative to non-coal regions, likely driven by falling population and income levels, which weaken local demand linkages. I also observe modest service employment reductions in lignite regions starting around 2007, likely reflecting the gradual reduction in mining employment and associated demand spillovers. In the agricultural sector (Panel 6d), I find little evidence of significant employment changes in either type of coal region relative to non-coal areas after 1961. In total employment (Panel 6a), hard coal regions exhibit a persistent decline after the phase-out. Lignite regions also show a downward trend, driven by service sector reductions, though the magnitude is considerably smaller.

To assess the magnitude of the long-run effects of the hard coal phase-out, Table A8 reports results from a difference-in-differences regression using manufacturing employment as the outcome. The specification replaces the event-time indicators with a post-1961 indicator. The estimates

³⁹Figure A14 in the Appendix shows that the results are robust to including all available post-2007 years (2007–2024), confirming that the choice of reference points does not drive the findings.

indicate that manufacturing employment in hard coal regions declined by 0.19 (log) sd relative to non-coal regions. In levels, this corresponds to an average decline of approximately 18%, suggesting that the phase-out had substantial and persistent effects on local industrial employment.⁴⁰

Taken together, these findings show that the economic benefits of hard coal production eroded as mining activities declined. Rather than leaving behind a lasting industrial advantage, former producer regions experienced a relative contraction in employment in manufacturing and service.

6.4 Understanding the Divergent Effects of Hard Coal and Lignite Mining

Heterogeneity by Production Linkages. To better understand the mechanisms behind the divergent economic effects of hard coal and lignite documented in the previous section, I disaggregate the manufacturing sector by the degree of upstream or downstream linkage to coal mining (see Section 4.1). For clarity and comparability, I estimate a difference-in-difference (DiD) specification based on Equation 1, in which the event-time indicators are replaced by a single post-period indicator. This enables a direct comparison of average employment changes across industries with varying degrees of production linkages to mining.

The left panel of Figure 7 shows estimates for the extraction period (1849–1939). In hard coal regions, manufacturing employment rose significantly after the onset of large-scale coal mining. Employment gains were more pronounced in industries with stronger input–output linkages to coal. For non-linked industries, I find smaller and statistically insignificant employment effects. By contrast, lignite regions experienced overall declines in manufacturing employment, with slightly smaller losses in linked industries. This pattern suggests that lignite mining did not generate substantial manufacturing linkages and was instead associated with local industrial decline. Disaggregating further, the positive employment effects in hard coal regions are observed across all types of linked industries, those connected to mining through upstream linkages, downstream linkages, or both. This suggests that hard coal mining supported both local input suppliers and coal-using industries. In lignite regions, by contrast, employment effects remain weak or negative across all linkage types. For industries linked both upstream and downstream, cost pressures and potential spillovers appear to offset each other, resulting in no net gain. For those with only one type

⁴⁰As discussed in Section 2.1, the lignite phase-out did not begin until the early 1990s and remains ongoing as of 2025. Therefore, the lignite coefficient in Table A8 should not be interpreted as the effect of a completed phase-out, but rather as reflecting average post-1961 employment dynamics relative to the pre-period.

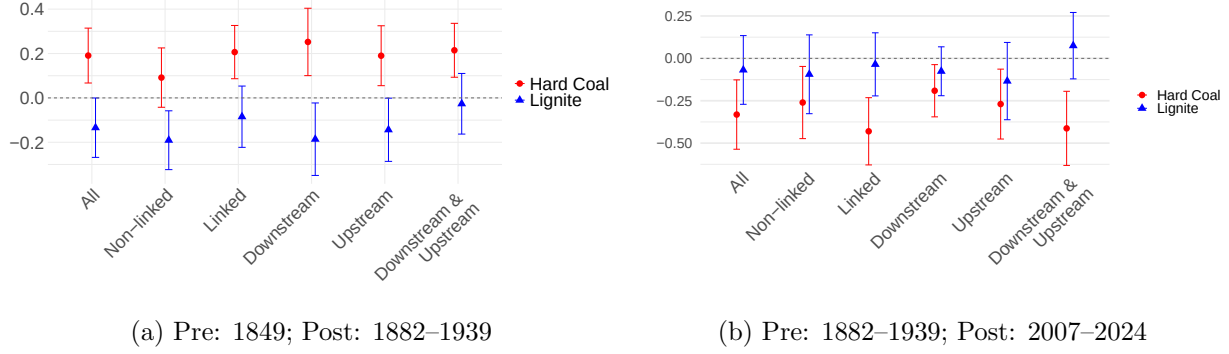


Figure 7: Difference-in-Differences regressions: Effect of Coal Endowment on Disaggregated Manufacturing Employment

Note: This figure plots coefficients and 95% confidence intervals from difference-in-differences regressions based on Equation 1, where the event-time indicators are replaced by a single post-period indicator. Both panels report estimated effects of hard coal (red) and lignite (blue) endowments on county-level manufacturing employment, disaggregated by linkage to coal mining. All outcomes are log-transformed and standardized to have a mean of zero and a sd of one. The left panel covers the extraction period using counties within Prussian borders (Pre: 1849; Post: 1882–1939). The post-period indicator equals one for all years after 1849. The right panel shows results for the coal phase-out period in West Germany (Pre: 1882–1939; Post: 2007–2024). Standard errors are clustered at the county level.

of linkage, negative effects dominate, consistent with crowding-out from rising wages and limited spillovers.

When turning to the disaggregated estimates of the shift–share approach in Figure A18a, a broadly similar pattern emerges: In hard coal regions, global price shocks are associated with positive employment responses in manufacturing sectors linked to coal, while effects in non-linked industries are negative (marginally significant). In lignite regions, by contrast, there is no meaningful employment response to coal price shocks across either sector type.

The right panel of Figure 7 presents estimates for the coal phase-out period (1882–2024).⁴¹ Here, the earlier patterns reverse. In hard coal regions, the sharpest employment declines occur in industries that had previously been linked to mining, especially those with both upstream and downstream linkages. These industries, tightly integrated into the coal-based economy, appear particularly vulnerable to the sector’s contraction. Non-linked industries also experienced significant, though smaller, declines. In lignite regions, by contrast, there are no significant differences across

⁴¹The specification defines the pre-period using benchmark years from 1882 to 1939, and the post-period as 2007–2024. Due to data limitations, disaggregated manufacturing employment by industry is unavailable between 1950 and 1987. As a result, the coefficient should be interpreted as the effect of the late phase-out period (post-2007), rather than the full phase-out timeline.

industry groups. This is consistent with lignite’s weaker integration into the surrounding industries.

Sectoral Concentration of Manufacturing Employment. A potential alternative explanation for the sharp decline in hard coal regions after the phase-out is that these areas had developed narrow, undiversified manufacturing structures, leaving them exposed to sector-specific shocks. To assess this, I examine changes in local industry concentration using the Herfindahl–Hirschman Index (HHI) of manufacturing employment (see Section 4.1). As shown in Figure A13, the opposite pattern emerges: during the extraction period, hard coal regions became more diversified across manufacturing industries than non-coal areas. This likely reflects the broad input–output linkages of hard coal mining, which supported a wide range of supplier and user industries. By contrast, lignite regions saw a slight increase in industrial concentration over time, likely driven by crowding-out effects that constrained the development of a more diverse industrial base. These results reinforce the interpretation that the vulnerability of hard coal regions was not driven by a narrow industrial structure, but by dependence on a single leading sector. It was the depth of integration with coal, rather sectoral concentration of manufacturing employment, that amplified long-run exposure to structural decline.

Taken together, my findings show that hard coal mining spurred local industrial growth through strong production linkages, while lignite mining generated few spillovers and contributed to deindustrialization. Yet the same linkages that supported growth in hard coal regions also deepened their exposure to decline, amplifying employment losses as mining contracted.

7 Robustness of the Results

To demonstrate that my results are not sensitive to the specific choice of variables, samples, or models, I perform several robustness checks.

Balanced Subsamples. A potential concern is that changes in county boundaries over time could bias results if sample composition is systematically related to coal endowments. To address this, I re-estimate the event-study specifications (Equation 1) using a balanced panel of counties consistently observed across the full time period. Figure A5 presents results for the extraction period (1849–1939). While standard errors are somewhat larger, the direction and magnitude of

the effects remain consistent with the baseline estimates from Figure 4. Figure A6 shows results for the coal phase-out period (1882–2024) using a balanced sample. Again, the patterns closely mirror the baseline estimates from Figure 6. Overall, these findings confirm that the main results are not driven by sample composition or administrative boundary changes.

Potential Confounding Factors. Table A11 reports results from a series of difference-in-differences regressions based on Equation 1 for the coal extraction period (1849–1939), using manufacturing employment as the outcome. The specification replaces event-time indicators with a single post-period indicator, and adds additional controls sequentially to assess the robustness of the estimated effects. Column (2) adds distance to Wittenberg, a commonly used proxy for historical Protestant influence, linked to differences in human capital and institutional development (S. O. Becker & Woessmann, 2009). Column (3) includes distance to the Polish border to account for possible variation in labor supply and cross-border migration. Column (4) controls for log distance to Berlin, reflecting potential advantages from proximity to the capital such as access to political influence or public investment. Column (5) adds a west-of-Elbe indicator to capture Germany’s historical east–west divide in agrarian structure and labor institutions (Tilly & Kopsidis, 2020). Finally, Column (6) includes a dummy for counties located in territories formerly occupied by France, accounting for long-run institutional effects of Napoleonic reforms (Acemoglu et al., 2011). Across all specifications, the estimated effects of coal exposure on manufacturing employment remain remarkably stable in both magnitude and statistical significance. This reinforces the interpretation that the main results are not confounded by pre-existing regional differences in geography, institutions, or human capital.

Potential Contamination Bias. To address concerns that including both coal types in a single regression may blur their distinct effects, I re-estimate the event-study model from Equation 1 separately for hard coal and lignite endowments. Figures A11 and A12 plot the coefficients from these separate specifications. The results closely mirror the baseline estimates reported in Figure 4, confirming that the observed effects are not driven by contamination across treatment types. As a further robustness check, I exclude all counties containing both hard coal and lignite deposits. Figure A8 presents event-study estimates based on this restricted sample. The results remain

consistent in both magnitude and significance, suggesting that the main findings are not driven by overlapping exposure.

Sensitivity to Outlier Regions. To ensure that the results are not driven by a disproportionately large coal-producing district, I re-estimate the difference-in-difference specifications excluding key outliers. Specifically, I drop the largest hard coal-producing district (Arnsberg) and the largest lignite-producing district (Merseburg), based on total production between 1850 and 1914 (own calculations using data from Kunz (2008)). As shown in Columns (2) and (4) of Table A12, the results remain robust in both magnitude and significance, indicating that the main findings are not driven by high-output districts.

Sensitivity to Timing of Extraction. Another potential concern is that the stronger economic effects observed in hard coal regions may reflect their earlier onset of extraction rather than structural differences in linkage intensity (see Figure A1a). To address this, I re-estimate the difference-in-difference specifications excluding Oppeln, the earliest major hard coal-producing district based on production levels in 1850 (own calculations using data from Kunz (2008)). As shown in Column (3) of Table A12, the results remain robust in both magnitude and statistical significance. For lignite mining, Merseburg was both the earliest and the largest producer over the period. Corresponding results excluding this district are reported in Column (4). These findings, together with the evidence that employment gains in hard coal regions were concentrated in sectors with strong upstream and downstream linkages, support the interpretation that the observed divergence reflects differences in economic integration, rather than differences in timing of extraction.

Assessing Potential Institutional Differences. To account for time-varying institutional differences across regions, I re-estimate the difference-in-difference specification including fixed effects for districts (*Regierungsbezirke*) interacted with a post-1849 indicator. This absorbs district-specific shocks or reforms that may have differentially affected hard coal, lignite, and non-coal regions after the onset of large-scale coal extraction. The results remain reported in Column (5) of Table A12 remain robust to this specification.

Alternative Definitions of Coal Exposure. A concern is that the estimated effects may be sensitive to how exposure to coal deposits is defined. In the baseline specification, I classify a county as exposed to hard coal or lignite if a deposit lies within a 20 kilometer radius of the county centroid. To test the robustness of the geographic treatment definition, I re-estimate the difference-in-differences specification based on Equation 1 for the coal extraction period (1849–1939), using manufacturing employment as the outcome and replacing event-time indicators with a single post-period indicator. I vary the distance threshold for coal endowment from the baseline 20 kilometers to alternative cutoffs of 10, 15, 25, and 30 kilometers. As shown in Table A10, the results remain qualitatively consistent across specifications. The baseline 20-kilometer radius neither maximizes the estimated effect of hard coal nor minimizes the negative effect of lignite. The estimated effect of hard coal exposure is statistically significant at distances up to 20 kilometers, but attenuates and becomes insignificant beyond that range. This spatial pattern suggests that the economic benefits of hard coal extraction were concentrated near production sites, consistent with the clustering of coal-linked industries and strong, localized input–output linkages. In contrast, the negative effects of lignite exposure remain statistically significant even at wider radii, suggesting that the adverse economic impacts of extraction extended into neighboring regions. Additionally, I restrict the analysis to coal deposits located within Prussia. The results, reported in Column (2) of Table A9, remain robust. Finally, I replace the binary treatment indicators with continuous measures of proximity to the nearest hard coal or lignite deposit, allowing treatment intensity to vary with distance. As shown in Column (3) of Table A9, the results align with the baseline: manufacturing employment increases with proximity to hard coal and decreases with proximity to lignite. Overall, these findings show that the main results are not driven by the specific geographic definition of coal exposure and hold across a range of alternative treatment measures.

Potential Spatial Correlation in Standard Errors. To account for potential spatial autocorrelation in the error structure, I re-estimate the baseline event-study model using Conley (1999) standard errors, which allow for correlation of residuals across geographic space. In addition, I follow S. Becker et al. (2025) and correct for spatial unit roots. The results, shown in Column (1) and (2) of Figure A9 using manufacturing employment as the outcome variable, remain marginally significant across most years.

Assessing Potential Pre-trends. The event-study model in Equation 1 relies on the assumption that, absent coal extraction, regions with and without deposits would have followed similar economic trends, conditional on fixed effects and observed controls. The discrete jump in employment after 1849, followed by stable post-treatment differences, already provides support to this assumption. To further assess its validity, I test for differential pre-trends using an alternative outcome: wages. Specifically, I re-estimate Equation 1 using unskilled wages as the outcome variable and control for local price levels to account for cost-of-living differences. The wage series is based on A.-K. Becker and Hornung (2025), who digitize unskilled wage data for 667 Prussian localities from 1800 to 1879 and extend the series to the county level through 1914 (see Section A.4 in the Appendix for details). Figure A10 plots the estimated effects of hard coal and lignite endowments on real wages from 1800 to 1914. The results show no large differences in wage levels or trends between coal and non-coal regions before 1849, supporting the parallel trends assumption. Starting in the mid-19th century, wages began to rise in hard coal regions, consistent with increasing labor demand driven by expanding mining and industrial activity. These patterns are consistent with demand-side effects from resource extraction and reinforce the interpretation that the observed employment trends are not driven by differential pre-treatment dynamics.

Alternative Time-Series Variation To assess the robustness of the shift–share estimates, I test whether the results are sensitive to alternative definitions of the time-varying shift component. In the baseline specification, the shift is given by the world coal price, which captures global demand and supply shocks in the coal market and is plausibly exogenous to local economic conditions in Germany. This helps address potential endogeneity concerns that may arise when using national variables, which could partially reflect local outcomes in coal-producing regions. As a robustness check, however, I re-estimate the shift–share model using coal-type-specific, rather than aggregate, indicators of coal sector activity. Specifically, I use several alternative national-level variables: German hard coal and lignite production, prices, and revenue (price times quantity).⁴² These

⁴²Using coal revenues as the shift variable is particularly informative, as it captures both supply-side movements (quantities) and demand-side movements (prices) that determine the mining sector’s overall scale. Quantity-only measures may understate differences between hard coal and lignite, since lignite has a lower unit value, even when extraction costs per unit of energy are similar. Conversely, price-only measures can be misleading, as prices may fluctuate due to supply disruptions, or trade restrictions, even when production remains stable. By combining both prices and quantities, production value offers a more comprehensive and economically meaningful proxy for the intensity of coal sector activity at the national level.

alternatives allow for a clearer differentiation between hard coal and lignite dynamics.

Moreover, I use a five-year moving average of the world coal price to account for potential lags in the adjustment of local economic conditions to global market shocks. As shown in Figure A17, the results remain consistent across specifications in both magnitude and significance. This confirms that the estimated effects are not sensitive to the precise choice of the shift variable.

To assess whether the estimated effects are driven by unrelated global trends rather than coal-specific price dynamics, I conduct placebo regressions in which global coal prices are replaced with prices of unrelated commodities (coffee, sugar, tobacco, and rice). The estimated interaction coefficients for hard-coal and lignite regions are near zero or slightly negative (Figure A18b), providing no evidence of analogous effects and reinforcing the validity of the identification strategy.

Alternative Input-Output Table and Linkage Definition. To ensure that the observed heterogeneity by linkage intensity is not driven by the specific input–output table used, I replicate the analysis using an alternative classification. Specifically, I use the historical input–output table from Fremdling and Staeglin (2014), which is based on 1936 data for Germany. This table enables me to disaggregate the industries in my occupation panel into 25 manufacturing sectors. Industries are reclassified by their upstream and downstream linkage intensity using this table (see Table A5). Then, I re-estimate the difference-in-differences models for both the coal extraction and the coal phase-out period. As shown in Figure A15 in the Appendix, the results remain highly consistent in both magnitude and significance, confirming that the findings are robust to the choice of input–output classification.

In addition, I assess whether the observed effects depend on the linkage definition. Figure A16 replicates the analysis using direct linkages only (left panel) and second-degree (indirect) linkages (right panel) for both upstream and downstream relationships, based on the U.S. 1947 input–output table (U.S. Bureau of Economic Analysis, 2019). The results remain robust to this alternative specification.

Together, these robustness checks confirm the validity of my results across both the extraction and the phase-out period, and reinforce the heterogeneity patterns.

8 Conclusion

This paper offers an explanation for why similar industries can lead regions down sharply divergent development paths, depending on the strength of local production linkages. My results suggest that industries are more likely to generate lasting local benefits when they create strong connections to the broader economy. Where an industry remains isolated, regions tend to stagnate or decline.

I compare two coal industries in Germany that operated under similar institutional conditions but differed in their integration into local supply chains. The results show that a resource sector's ability to generate upstream and downstream linkages is a key mechanism shaping its broader local economic impact.

My findings show that regions specialized in hard coal mining, an industry with strong input–output linkages, experienced significant employment gains during the extraction period, with spillovers into both manufacturing and services. Manufacturing gains were more pronounced in industries with upstream or downstream linkages to coal mining. Hard coal regions also became more diversified across manufacturing sub-sectors over time, indicating that resource-based growth did not lead to industrial concentration. However, after the coal phase-out, hard coal regions suffered persistent employment losses, especially in sectors tightly connected to mining. My results suggest that this vulnerability was not due to a lack of diversification across industries, but rather to the depth of integration with the coal economy, which amplified the costs of decline. In contrast, lignite regions, where extraction is more isolated from the local economy, saw adverse effects even during extraction, including deindustrialization and service sector decline. Yet their weaker integration into industrial supply chains made them less exposed to economic disruptions during the subsequent decline in lignite production. Taken together, my findings highlight the double-edged nature of production linkages: while they may amplify growth when a major industry expands, they may also amplify losses when it contracts. This mechanism helps explain why regions exposed to seemingly similar industries can experience sharply divergent development paths.

My findings contribute to current debates on place-based and industrial policy in advanced economies. Regional disparities have been widening in recent decades, raising concerns about political stability and social cohesion (Autor et al., 2020; Rodríguez-Pose, 2018). I show that targeting tradable sectors with high local linkage potential can generate larger local multipliers than support-

ing enclaved industries with limited integration into the regional economy. In some cases, support for such isolated sectors may even crowd out other activities and hinder broader development. My results suggest that designing place-based industrial policy around local production linkages can promote regional development and help narrow spatial disparities. At the same time, my results show that when a major industry begins to decline, the very linkages that once supported growth can transmit negative shocks across the local economy. Without effective transition strategies, regions with tightly embedded industrial structures may face deeper employment losses and more severe deindustrialization. This dynamic underscores the importance of forward-looking adjustment policies, particularly in the context of structural transformations such as decarbonization, where the phase-out of core industries is foreseeable. Moreover, this paper helps reconcile conflicting findings in the resource literature and sheds light on why the legacy of resource wealth can range from long-run prosperity to persistent underdevelopment.

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A Appendix

A.1 Additional Tables

A.1.1 Descriptives

Table A1: Descriptive Statistic: Cross-Sectional Data

Variable	N	Mean	SD	Min	Max
Panel (a): Treatment Variables					
Distance Hard coal (in km)	1122	132	111	0	552
Distance Lignite (in km)	1122	110	110	0	566
Hard coal region (dummy)	1122	0.16	0.36	0	1
Lignite region (dummy)	1122	0.18	0.39	0	1
Hard coal or lignite region (dummy)	1122	0.3	0.46	0	1
Hard coal and lignite region (dummy)	1122	0.03	0.18	0	1
Panel (b): Geographic Control Variables					
Distance River (in km)	1122	13	18	0	113
Distance Coast (in km)	1122	223	140	0	455
Caloric Suitability	1122	1667	274	714	2313
Slope	1122	1.2	1.1	0.03	12
Temperature	1122	8.5	0.84	5.8	11
Rainfall	1122	63	15	42	143

Note: This table reports summary statistics for key cross-sectional variables used in the main analyses. Units of observation are counties as of 1900. N refers to the number of counties.

Table A2: Descriptive Statistic: Pooled Panel Data

Variable	N	Mean	SD	Min	Max
Panel A: 1849–1939					
Total Employment	2365	31592	54052	3775	1774040
Manufacturing Employment	2365	10327	28129	415	908951
Service Employment	2365	7419	26494	163	849575
Agricultural Employment	2365	12686	8605	84	61594
Panel B: 1882–1939					
Total Employment	5100	23995	42357	677	1774040
Manufacturing Employment	5100	9056	21882	240	908951
Service Employment	5100	6193	20442	97	849575
Agricultural Employment	5100	8169	5549	15	38107
Panel C: 1882–2024					
Total Employment	7642	29583	45025	768	1774040
Manufacturing Employment	7642	11896	21208	385	908951
Service Employment	7642	12367	26731	123	849575
Agricultural Employment	7642	4804	4807	0	32285

Note: This table reports summary statistics for key panel variables used in the main analyses. Panel (a) covers all Prussian counties from 1849 to 1939. Panel (b) covers counties within the borders of the German Empire from 1882 to 1939 (balanced panel). Panel (c) covers counties in West Germany from 1882 to 2024. N refers to the number of observations.

Table A3: Descriptive Statistic: Time-Series Data (1882–1939)

Variable	N	Mean	SD	Min	Max
World Coal Price	5100	93	7.3	83	104
World Coal Price (5Y Avg.)	5100	99	11	86	116
Hard Coal Production	5100	109	48	46	186
Lignite Production	5100	65	48	12	141
Hard Coal Price	5100	9.6	3.5	5.1	15
Lignite Price	5100	2.6	0.24	2.4	3
Hard Coal Revenue	5100	1148	705	236	2130
Lignite Revenue	5100	165	119	36	335

Note: This table reports summary statistics for key time-series variables used in the main analysis, based on counties in the German Empire from 1882 to 1939. N refers to the number of observations.

Table A4: Upstream and Downstream Linkage Measures for U.S. Input–Output Sectors (1947)

Industry	Upstream		Downstream		Linked		
	Direct	Indirect	Direct	Indirect	Downstream	Upstream	Total
Apparel and leather and allied products	0.0	0.2	0.1	0.6	No	No	No
Chemical products	1.5	0.4	1.8	1.2	No	Yes	Yes
Fabricated metal products	1.0	0.7	1.7	5.8	Yes	Yes	Yes
Food and beverage and tobacco products	0.0	0.2	0.1	0.6	No	No	No
Machinery	2.9	0.7	0.3	3.7	Yes	Yes	Yes
Miscellaneous manufacturing	0.1	0.6	2.9	2.6	Yes	No	Yes
Motor vehicles, bodies and trailers, and parts	0.7	0.4	0.2	2.9	Yes	No	No
Nonmetallic mineral products	0.9	0.5	15.2	1.8	Yes	Yes	Yes
Paper products	0.3	0.4	1.9	1.0	No	No	Yes
Petroleum and coal products	1.2	0.8	5.1	2.4	Yes	Yes	Yes
Plastics and rubber products	0.2	0.9	0.7	1.2	No	No	No
Primary metals	0.8	0.7	14.4	1.1	Yes	Yes	Yes
Printing and related support activities	0.3	0.6	0.1	1.5	No	No	No
Textile mills and textile product mills	0.0	0.4	0.3	0.5	No	No	No
Wood products	2.1	0.4	0.4	1.6	No	Yes	Yes

Note: This table reports direct and indirect upstream (sales to mining) and downstream (inputs from mining) linkage shares in percent for U.S. manufacturing and service sectors, based on the 1947 summary-level Use table (before redefinitions) (U.S. Bureau of Economic Analysis, 2019). I compute linkage shares on the full I-O matrix, but report only those industries that can be matched to historical occupations, and I exclude own-use (diagonal). Consequently, the reported column sums for upstream and downstream shares (direct and indirect) need not equal one. Binary “linked” indicators are defined as specified in Section 4.1.

Table A5: Upstream and Downstream Linkage Measures for German Input–Output Sectors (1936)

Industry	Upstream		Downstream		Linked		
	Direct	Indirect	Direct	Indirect	Downstream	Upstream	Total
Building and construction	0.3	1.3	0.4	2.8	No	Yes	Yes
Chemical industry	2.4	0.4	4.6	5.2	Yes	Yes	Yes
Chemical-technical industry	0.0	1.2	0.0	4.1	No	No	Yes
Clothing	0.0	1.3	0.0	1.1	No	Yes	Yes
Edible oil and fats	0.0	0.1	0.0	2.0	No	No	No
Electrical engineering	0.4	1.0	0.5	4.0	Yes	Yes	Yes
Electricity, gas and water	4.0	0.7	33.4	3.0	Yes	Yes	Yes
Fabricated iron and steel products	0.6	0.9	0.0	6.8	Yes	Yes	Yes
Food, beverages and tobacco	0.0	0.7	0.1	2.4	No	No	Yes
Foundries	6.4	0.4	0.0	6.8	Yes	Yes	Yes
Fuel industries	0.0	1.2	18.8	2.6	Yes	No	Yes
Glass	0.0	0.6	0.0	10.2	Yes	No	No
Leather industry	0.0	0.2	0.0	2.0	No	No	No
Machinery	0.0	1.6	0.0	4.4	No	Yes	Yes
Manufacture of paper and paper products	0.0	0.4	0.0	5.2	Yes	No	No
Manufactured wood products	0.2	0.6	0.1	2.5	No	No	Yes
Metal products	0.0	1.5	0.0	5.5	Yes	Yes	Yes
Non-ferrous metals	0.9	0.5	4.8	4.1	Yes	Yes	Yes
Precision engineering, optics	0.0	1.2	0.0	3.5	No	No	No
Printing and duplicating	0.4	0.8	0.0	1.8	No	No	No
Rubber and asbestos manufacturing	0.0	1.4	0.0	5.9	Yes	Yes	Yes
Saw mills, timber processing	12.9	0.2	0.0	2.8	No	Yes	Yes
Spirits industry	0.0	0.5	0.0	1.8	No	No	No
Stone and quarrying	0.1	0.8	3.5	13.7	Yes	No	Yes
Textiles	0.1	0.1	0.0	2.8	No	No	No

Note: This table reports direct and indirect upstream (sales to mining) and downstream (inputs from mining) linkage shares in percent for German manufacturing sectors, based on the 1936 I-O table (Fremdling & Staeglin, 2014). I compute linkage shares on the full I-O matrix, but report only those industries that can be matched to historical occupations, and I exclude own-use (diagonal). Consequently, the reported column sums for upstream and downstream shares (direct and indirect) need not equal one. Binary “linked” indicators are defined as specified in Section 4.1.

A.1.2 Event-studies

Table A6: Event-Study: Effect of Resource Endowment on Sectoral Employment (1849–1939)

	Total (1)	Manufacturing (2)	Service (3)	Agricultural (4)
HardCoal x 1882	0.399*** (0.084)	0.207*** (0.059)	0.126** (0.064)	0.089 (0.083)
HardCoal x 1895	0.466*** (0.105)	0.246*** (0.072)	0.157** (0.073)	0.085 (0.077)
HardCoal x 1907	0.468*** (0.119)	0.268*** (0.082)	0.188** (0.081)	-0.033 (0.096)
HardCoal x 1925	0.396*** (0.142)	0.185* (0.098)	0.127 (0.085)	-0.087 (0.096)
HardCoal x 1939	0.439*** (0.160)	0.166 (0.104)	0.154 (0.095)	-0.089 (0.092)
Lignite x 1882	-0.086 (0.093)	-0.073 (0.058)	-0.129** (0.059)	0.342*** (0.091)
Lignite x 1895	-0.242** (0.111)	-0.214*** (0.076)	-0.181** (0.071)	0.312*** (0.084)
Lignite x 1907	-0.343*** (0.127)	-0.283*** (0.091)	-0.245*** (0.080)	0.268** (0.108)
Lignite x 1925	-0.371*** (0.140)	-0.269*** (0.102)	-0.232*** (0.080)	0.290*** (0.107)
Lignite x 1939	-0.364** (0.166)	-0.192* (0.111)	-0.192** (0.092)	0.192* (0.113)
Geo Controls	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	2,365	2,365	2,365	2,365
Adjusted R ²	0.72	0.86	0.88	0.82

Notes: This table reports coefficients from event-study regressions based on Equation 1. The dependent variables are the log of standardized county-level employment in (1) total, (2) manufacturing, (3) service, and (4) agricultural sectors (mean zero, unit variance). Each row presents estimated effects for counties with hard coal or lignite deposits interacted with year fixed effects. the omitted year is 1849. All regressions include geographic controls, county fixed effects, and year fixed effects. Standard errors, clustered at the county level, are shown in parentheses. The analysis is restricted to Prussian counties. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.1.3 Difference-in-Difference Regressions

Table A7: Difference-in-Difference Regressions: Effect of Resource Endowment on Sectoral Employment (1849–1939)

	Total (1)	Manufacturing (2)	Service (3)	Agricultural (4)
HardCoal x Post1849	0.423*** (0.111)	0.208*** (0.074)	0.143** (0.073)	-0.008 (0.076)
Lignite x Post1849	-0.300*** (0.114)	-0.217*** (0.079)	-0.209*** (0.068)	0.275*** (0.087)
Geo Controls	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	2,365	2,365	2,365	2,365
Adjusted R ²	0.69	0.85	0.87	0.82

Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, where the event-time indicators are replaced by a single post-period indicator. The dependent variables are the log of standardized county-level employment in (1) total, (2) manufacturing, (3) services, and (4) agriculture (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits relative to non-coal countries interacted with a post indicator equal to one for all years after 1849. All regressions include geographic controls interacted with year fixed effects, county and year fixed effects. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8: Difference-in-Difference Regressions: Effect of Coal Phase-Out on Sectoral Employment (1882–2024)

	Total (1)	Manufacturing (2)	Service (3)	Agricultural (4)
HardCoal x Post1961	-0.146* (0.075)	-0.192*** (0.073)	-0.124** (0.053)	-0.112** (0.044)
Lignite x Post1961	-0.116* (0.062)	-0.024 (0.069)	0.010 (0.048)	-0.050 (0.045)
Geo Controls	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	7,642	7,642	7,642	7,642
Adjusted R ²	0.82	0.83	0.90	0.91

Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, where the event-time indicators are replaced by a single post-period indicator. The dependent variables are the log of standardized county-level employment in (1) total, (2) manufacturing, (3) services, and (4) agriculture (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits relative to non-coal countries interacted with a post indicator. The post indicator equals one for all years after 1961. All regressions include geographic controls interacted with year fixed effects, county and year fixed effects. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to West German counties. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A9: Difference-in-Difference Regressions: Alternative Definitions of the Treatment Variable (1849–1939)

	Baseline	Coal Inside Prussia	Proximity
	(1)	(2)	(3)
HardCoal x Post1849	0.208*** (0.074)	0.185** (0.074)	0.074* (0.042)
Lignite x Post1849	-0.217*** (0.079)	-0.206*** (0.079)	-0.158*** (0.030)
Geo Controls	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	2,365	2,365	2,365
Adjusted R ²	0.85	0.85	0.85

Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, where the event-time indicators are replaced by a single post-period indicator. The dependent variable is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each row shows estimated effects for counties classified as exposed to hard coal or lignite, with exposure defined in different ways, interacted with a post indicator equal to one for all years after 1849. All regressions include geographic controls, as well as county and year fixed effects. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A10: Difference-in-Difference Regressions: Varying the Distance Cut-off (1849–1939)

	10km (1)	15km (2)	20km (3)	25km (4)	30km (5)
HardCoal x Post1849	0.242*** (0.067)	0.136* (0.070)	0.208*** (0.074)	0.117 (0.073)	0.042 (0.074)
Lignite x Post1849	-0.231*** (0.079)	-0.246*** (0.077)	-0.217*** (0.079)	-0.284*** (0.081)	-0.291*** (0.076)
Geo Controls	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	2,365	2,365	2,365	2,365	2,365
Adjusted R ²	0.85	0.85	0.85	0.85	0.85

Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, where the event-time indicators are replaced by a single post-period indicator. The dependent variables is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits, defined using varying distance cutoffs, interacted with a post indicator equal to one for all years after 1849. All regressions include geographic controls, county and year fixed effects. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A11: Difference-in-Difference Regressions: Potential Confounders (1849–1939)

	Baseline (1)	Distance Wittenberg (2)	Distance Poland (3)	Distance Berlin (4)	West Elbe (5)	French Occupation (6)
HardCoal x Post1849	0.208*** (0.074)	0.117* (0.068)	0.121* (0.068)	0.161** (0.068)	0.188*** (0.067)	0.190*** (0.067)
Lignite x Post1849	-0.217*** (0.079)	-0.153* (0.092)	-0.160* (0.092)	-0.101 (0.079)	-0.163** (0.081)	-0.161** (0.081)
Geo Controls	Yes	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,365	2,365	2,365	2,365	2,365	2,365
Adjusted R ²	0.85	0.86	0.86	0.86	0.86	0.86

Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, where the event-time indicators are replaced by a single post-period indicator. The dependent variables is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits relative to non-coal countries interacted with a post indicator equal to one for all years after 1849. All regressions include geographic controls, county and year fixed effects. Potential confounders are added one at a time. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A12: Difference-in-Difference Regressions: Specification Robustness (1849–1939)

	Baseline (1)	Largest Producer Hard coal (2)	First Mover Hard Coal (3)	First Mover = Largest Producer Lignite (4)	District x Post1849 (5)
HardCoal x Post1849	0.208*** (0.074)	0.131* (0.076)	0.151** (0.076)	0.163** (0.080)	0.218*** (0.072)
Lignite x Post1849	-0.217*** (0.079)	-0.196** (0.081)	-0.199** (0.079)	-0.254*** (0.085)	-0.339*** (0.103)
Geo Controls	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	2,365	2,227	2,230	2,257	2,365
Adjusted R ²	0.85	0.85	0.86	0.85	0.86

Notes: This table reports coefficients from difference-in-differences regressions based on Equation 1, where the event-time indicators are replaced by a single post-period indicator. The dependent variable is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits relative to non-coal counties interacted with a post indicator equal to one for all years after 1849. All regressions include geographic controls, county and year fixed effects. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.2 Additional Figures

A.2.1 Descriptives

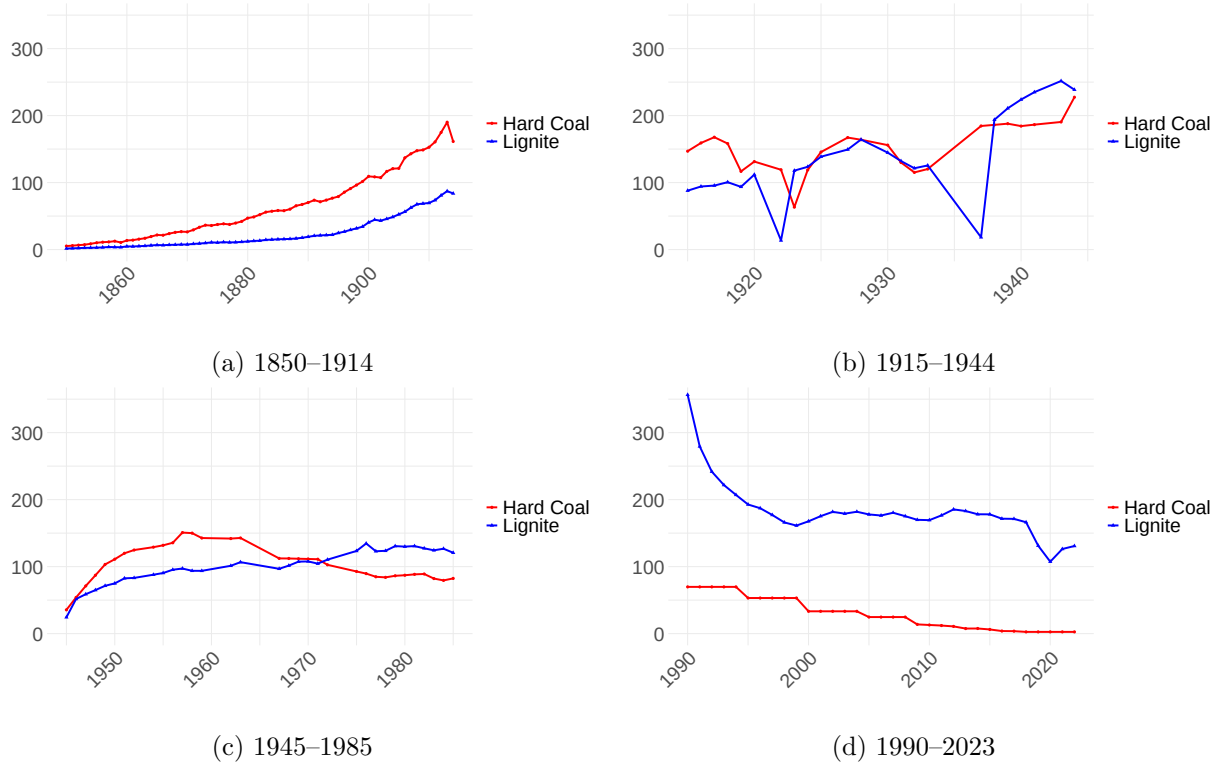


Figure A1: Hard Coal and Lignite Production over Time in Germany (1850–2023)

Note: This figure shows hard coal and lignite production in Germany over time, measured in million tons. Data for the period 1850–1985 are from Fischer (2011); data for 1990–2023 are from (Statista Research Department, 2025a, 2025b).

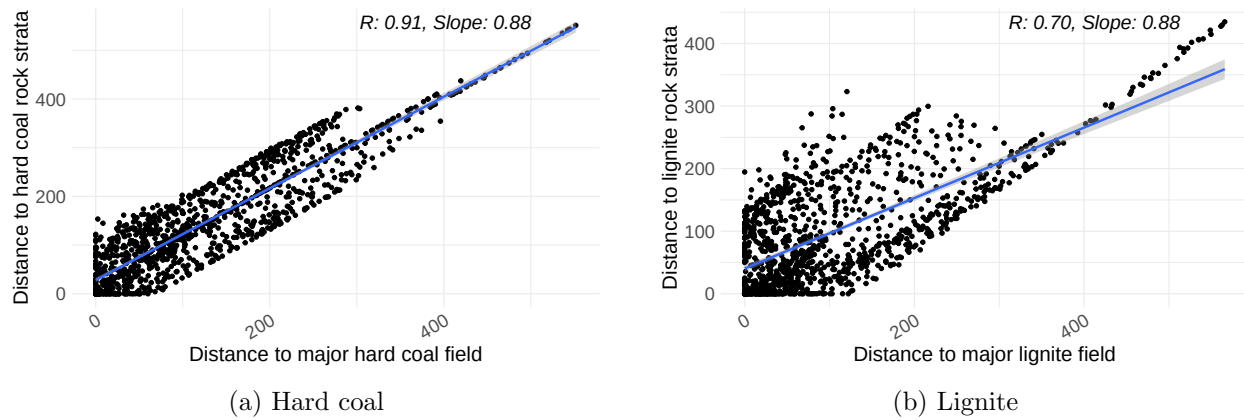


Figure A2: Correlation Between Distance to Major Coal Fields and Distance to Rock Strata in Germany

Note: The scatterplots show the correlation between the centroid distance of a county to the nearest large hard coal (left) or lignite (right) field and to the nearest carboniferous or lignite rock strata, respectively (in km). Data sources: Asch and Bellenberg, 2005; Fernihough and O'Rourke, 2021.

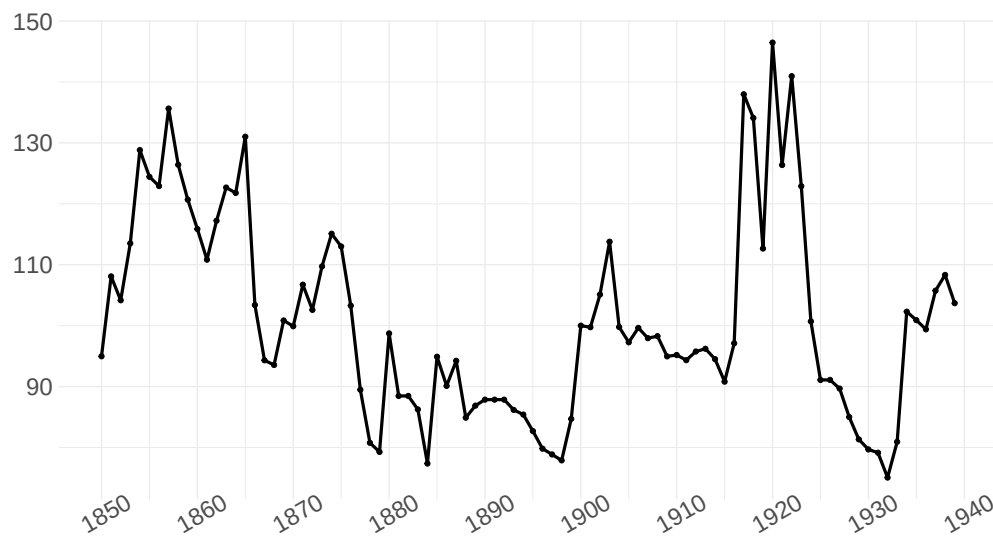


Figure A3: World Coal Prices over Time

Note: The figure plots annual world coal prices, 1850–1939, expressed in 1900 U.S. dollars. Source: Jacks (2019).



(a) Lignite Mining (1900, Senftenberg)
© Deutsche Fotothek (Franz Stoedtner)



(b) Hard Coal Mining (1893, Freital-Burgk)
© Deutsche Fotothek (Heinrich Börner)

Figure A4: Illustrating Differences in Coal Extraction Methods: Hard coal and Lignite Mining

Note: Panel (a) shows a typical lignite mine in 1900, where extraction was carried out via open-pit surface mining. Panel (b) depicts hard coal mining in the late 19th century, which relied on complex underground extraction techniques, including shafts, tunnels, and ventilation systems.

A.2.2 Event-studies

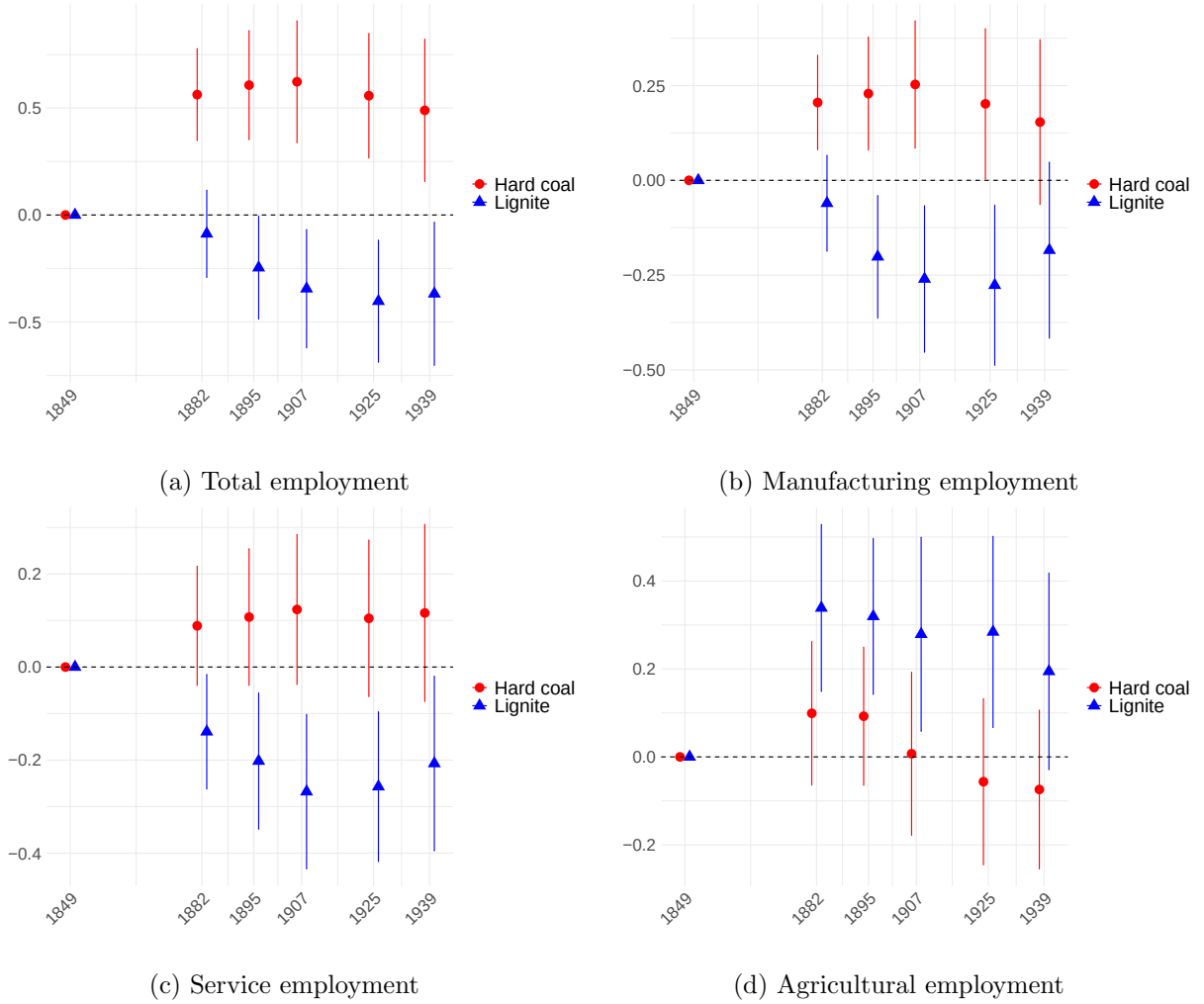


Figure A5: Event-Study: Effect of Resource Endowment on Sectoral Employment (1849–1939, Balanced Sample)

Note: This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1 using a balanced sample of Prussian counties. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment, (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite endowment indicators are interacted with year fixed effects; the omitted year is 1849. Standard errors are clustered at the county level.



Figure A6: Event-Study: Effect of Coal Phase-Out on Sectoral Employment (1882–2024, Balanced Sample)

Note: This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1 using a balanced sample. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment, (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite indicators are interacted with year fixed effects; the omitted year is 1961. Standard errors are clustered at the county level. The analysis is restricted to West German counties.

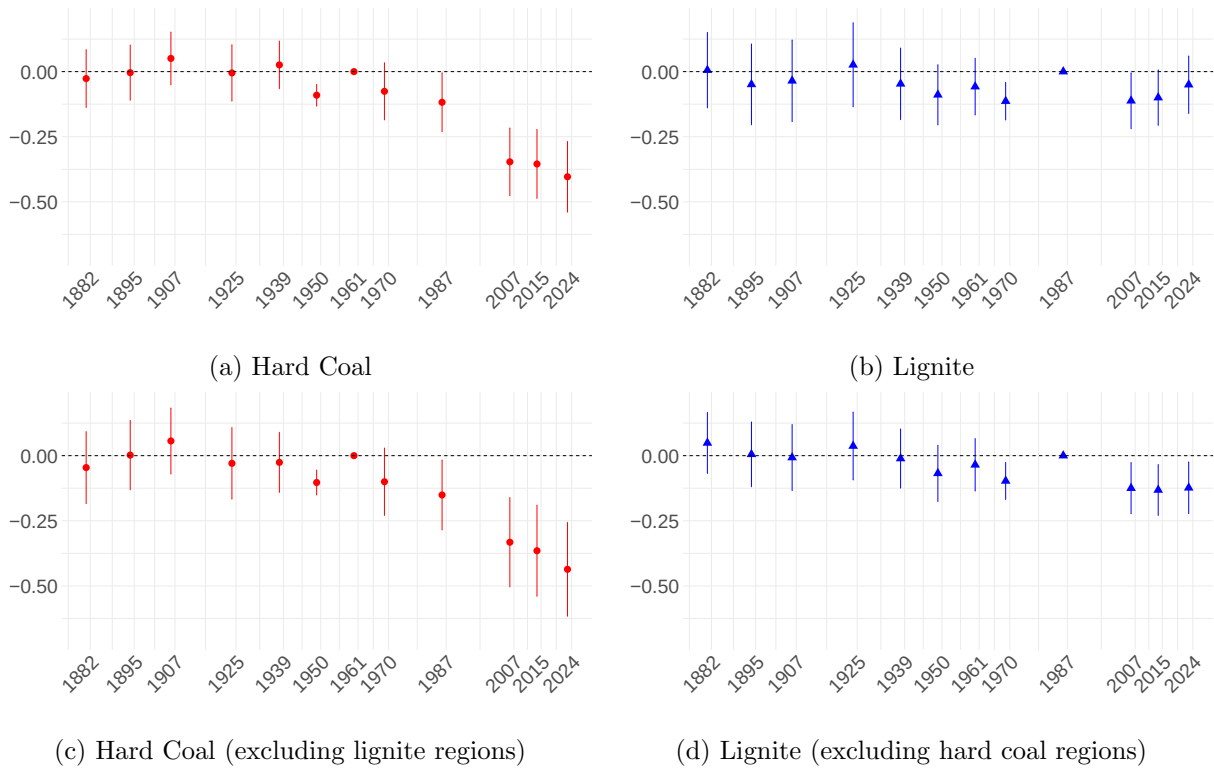


Figure A7: Event-Study: Effect of Coal Phase-Out on Manufacturing Employment (1882–2024, Separate Regressions)

Note: This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1. The dependent variable is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each specification interacts either the hard coal or lignite indicator with year fixed effects. The omitted base year is 1961 for hard coal and 1897 for lignite. Panels (c) and (d) exclude lignite and hard coal regions, respectively, to isolate effects by coal type. Standard errors are clustered at the county level. The sample is restricted to West German counties.

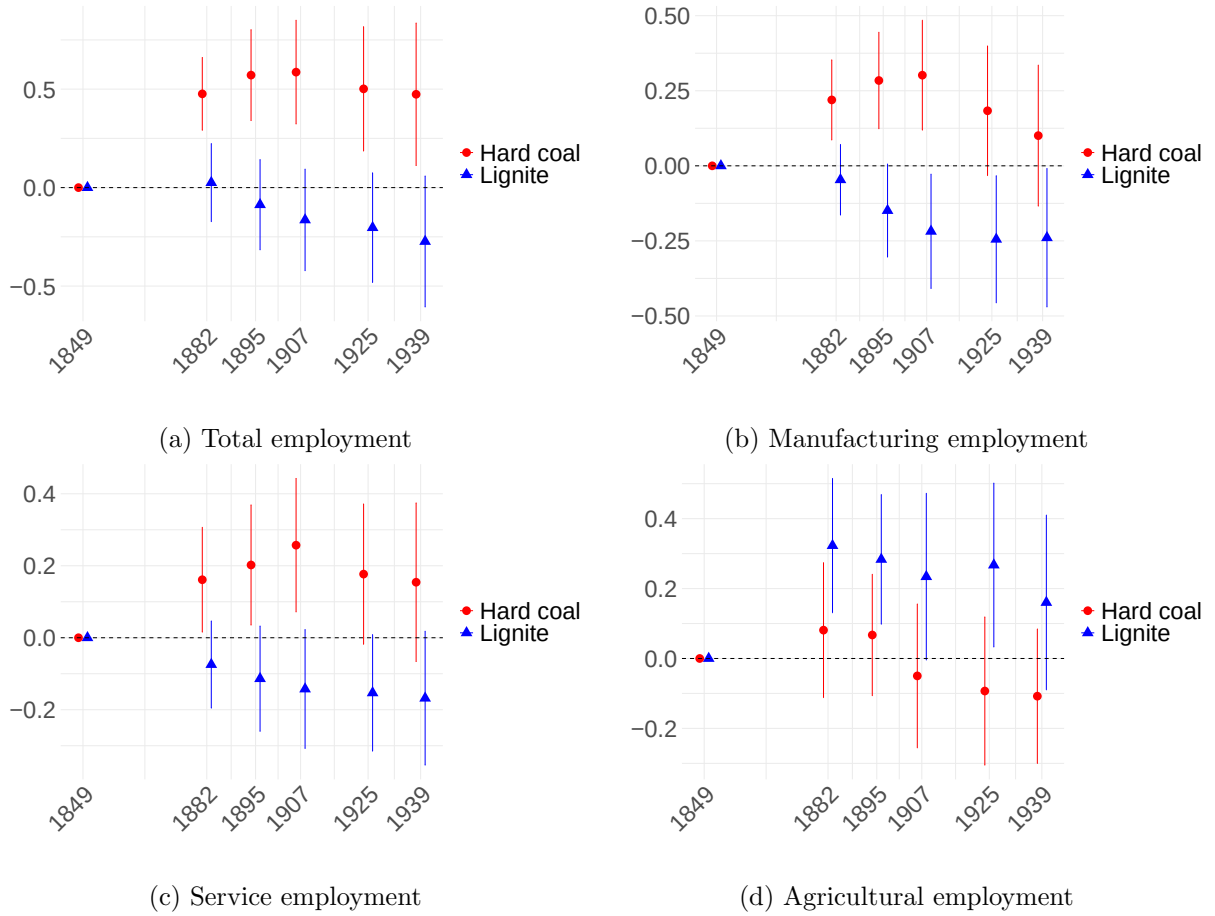


Figure A8: Event-Study: Effect of Coal Endowments on Sectoral Employment (1849–1939, Excluding Overlapping Deposits)

Note: This figure plots coefficient estimates and 95% confidence intervals from event-study regressions based on Equation 1. Counties with overlapping deposits of hard coal and lignite are excluded from the analysis. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment, (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite indicators are interacted with year fixed effects; the omitted year is 1849. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties.

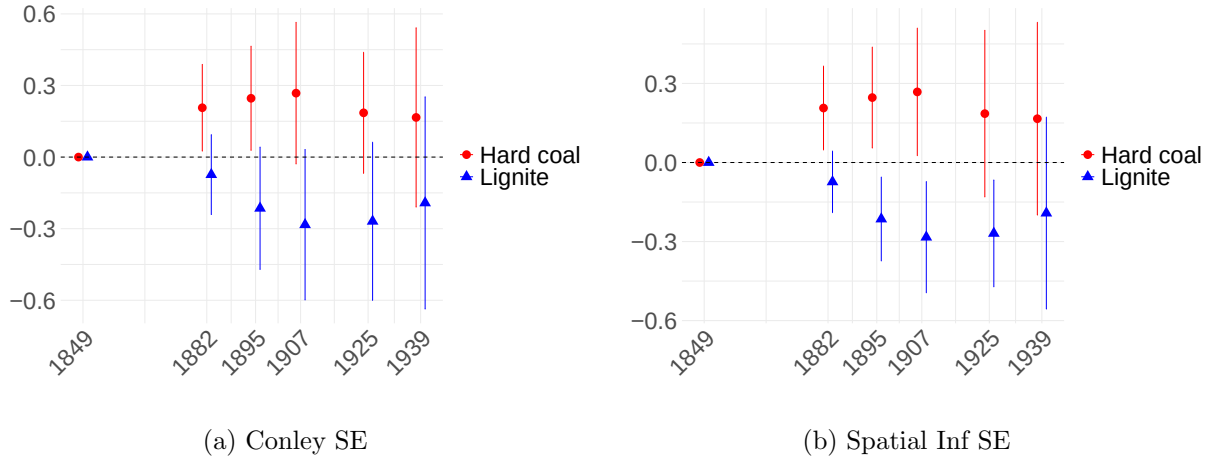


Figure A9: Event-study: Effect of Coal Endowments on Manufacturing Employment (1849–1939, Alternative Inference Methods)

This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1. Panel (a) reports Conley standard errors with a 100 km cut-off (Conley, 1999). Panel (b) implements the spatial inference method proposed by S. Becker et al. (2025), which adjusts for spatial unit roots. The dependent variable is the log of standardized county-level employment in manufacturing (mean zero, unit variance). The estimated effects of hard coal (red) and lignite (blue) endowment are shown relative to the omitted baseline year 1849. Standard errors are clustered at the county level. The sample is restricted to Prussian counties.

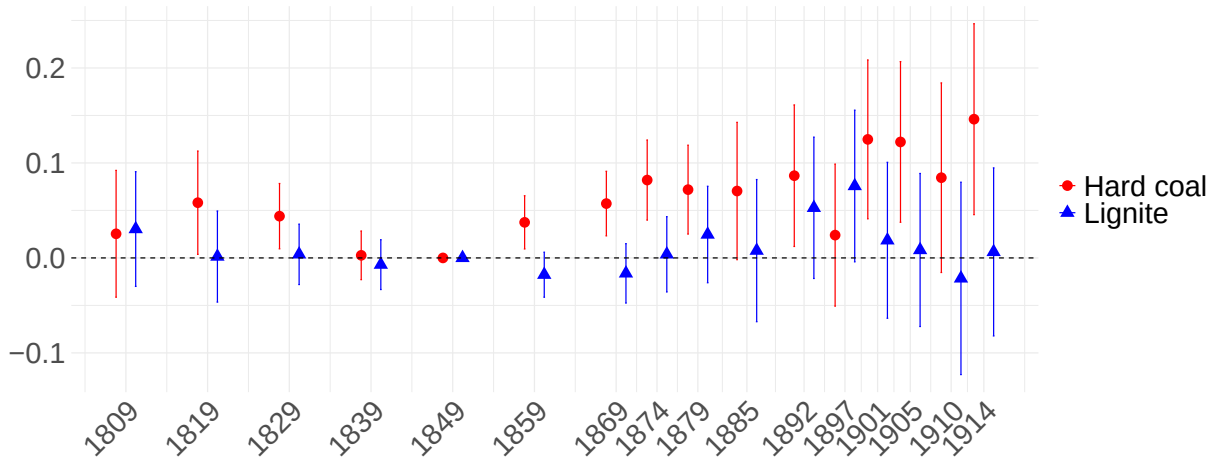


Figure A10: Event-study: Effect of Coal Endowments on Wages (1800–1914)

This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1, using unskilled wages as the outcome variable (see Section A.4 for details). The estimated effects of hard coal (red) and lignite (blue) endowment are shown relative to the omitted baseline year 1849. All regressions control for wheat prices in the nearest market, county fixed effects, and year fixed effects. Standard errors are clustered at the county level. The level of observation is the county, and the sample is restricted to Prussian counties.

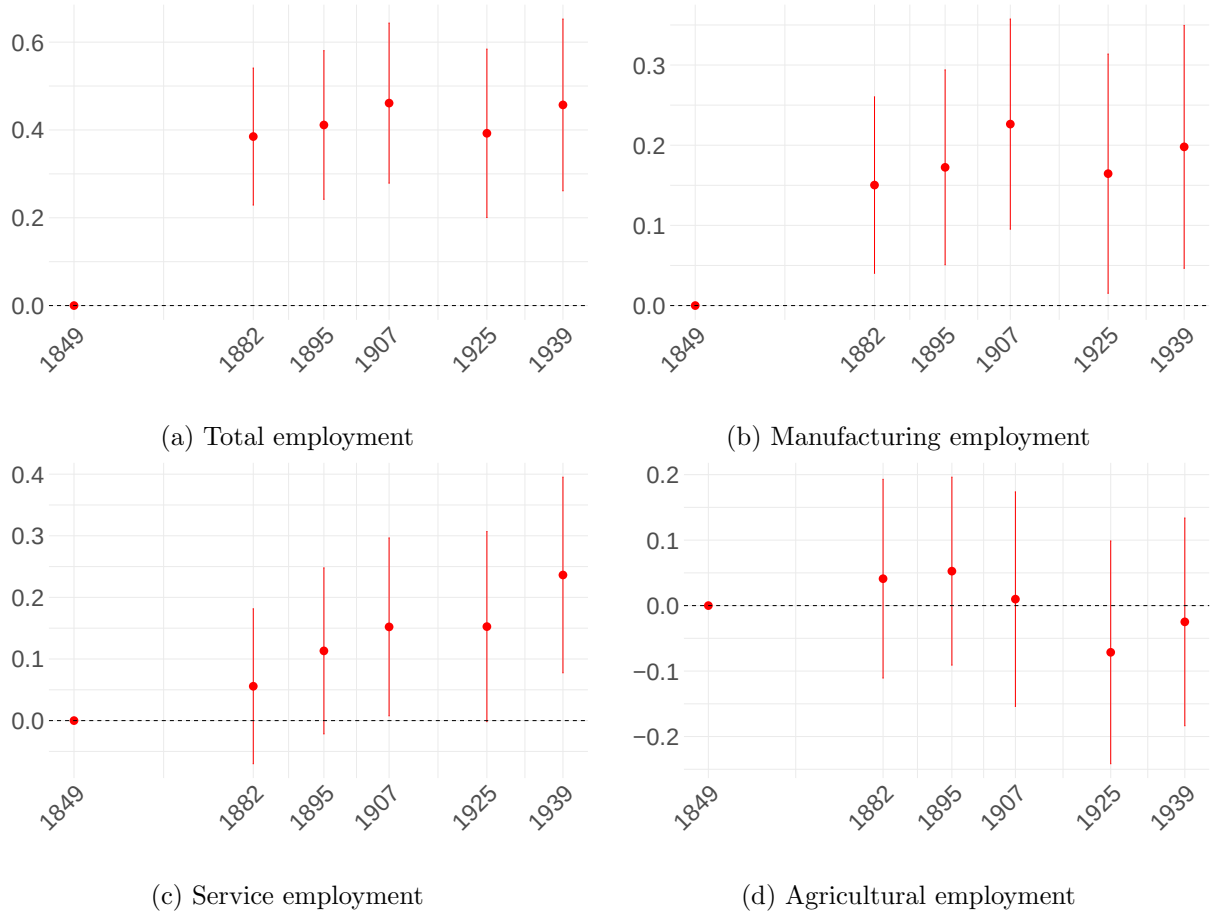


Figure A11: Event-Study: Effect of Hard Coal Endowment on Sectoral Employment (1849–1939)

Note: This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1. Each panel reports the estimated effects of hard coal exposure on county-level employment outcomes: (a) total employment, (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite endowment indicators are interacted with year fixed effects; the omitted year is 1849. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties.

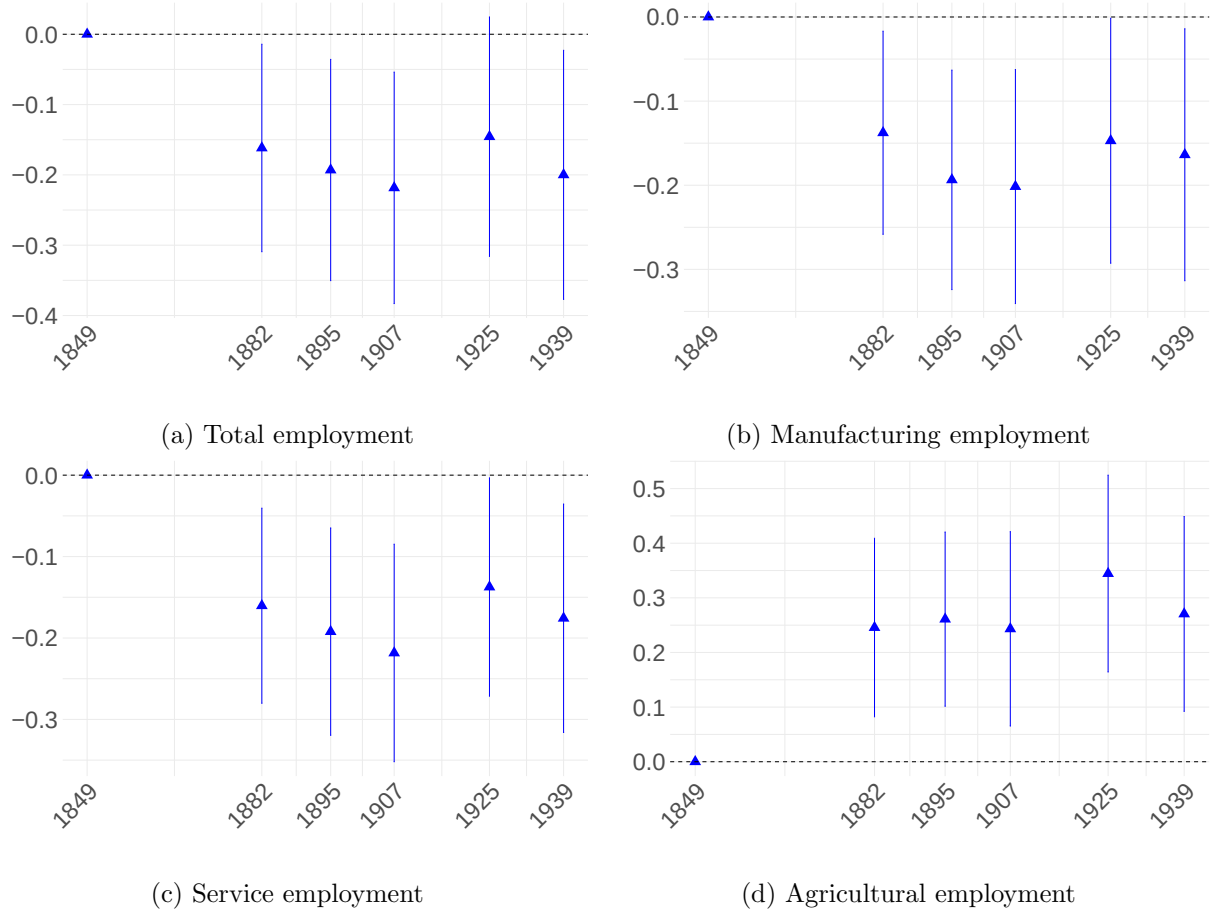


Figure A12: Event-Study: Effect of Hard Coal Endowment on Sectoral Employment (1849–1939)

Note: This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1. Each panel reports the estimated effects of exposure on county-level employment outcomes: (a) total employment, (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite endowment indicators are interacted with year fixed effects; the omitted year is 1849. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties.

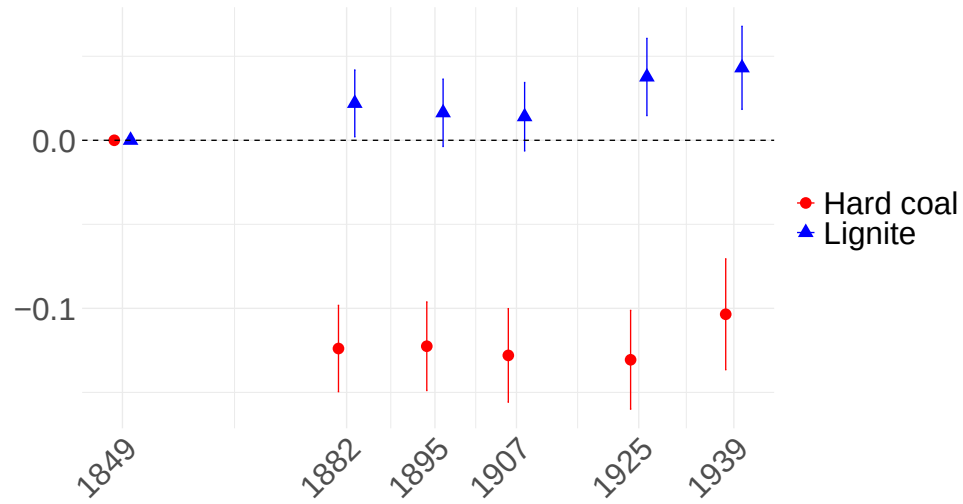


Figure A13: Event-Study: Effect of Coal Endowment on Sectoral Concentration of Manufacturing Employment (1849–1939).

Note: This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1, using the Herfindahl–Hirschman Index of manufacturing employment as the outcome variable. Higher index values reflect greater concentration, while lower values indicate a more diversified structure (see Section 4.1 for details). The estimated effects of hard coal (red) and lignite (blue) endowment are shown relative to the omitted baseline year 1849. The outcome is logged and standardized to have a mean of zero and a sd of one. Hard coal and lignite endowment indicators are interacted with year fixed effects; the omitted year is 1849. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties.

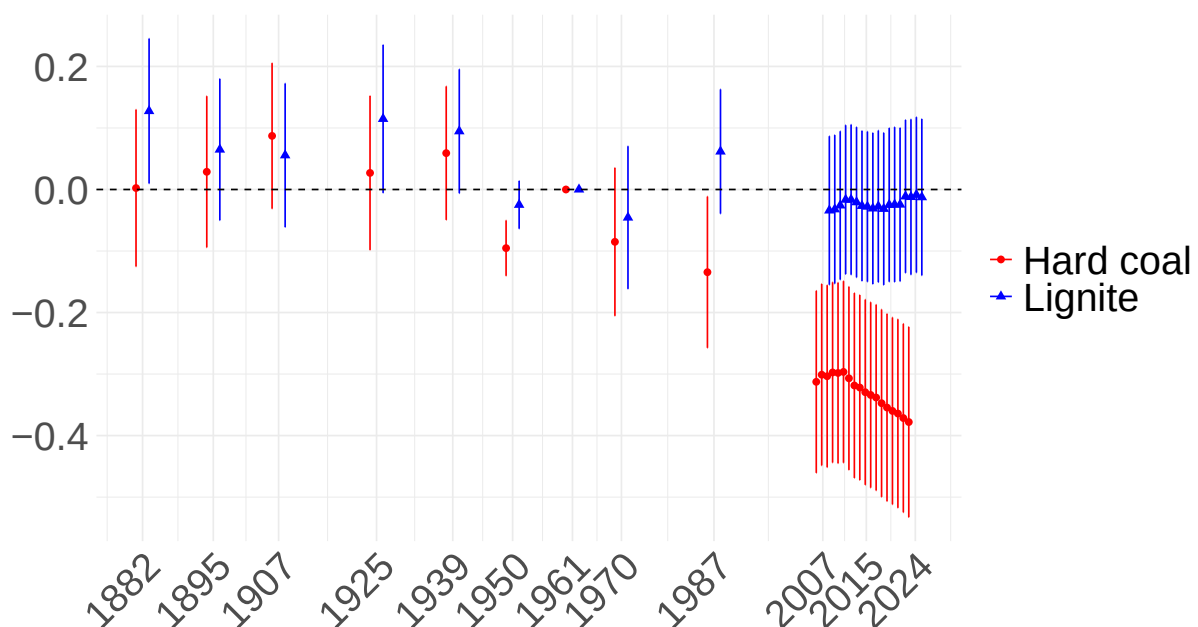


Figure A14: Event-Study: Effect of Coal Phase-Out on Sectoral Employment (1882–2024, all Post 2007 Years included)

Note: This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1. The dependent variable is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Hard coal and lignite indicators are interacted with year fixed effects; the omitted year is 1961. Standard errors are clustered at the county level. The analysis is restricted to West German counties.

A.2.3 Difference-in-Difference Regressions

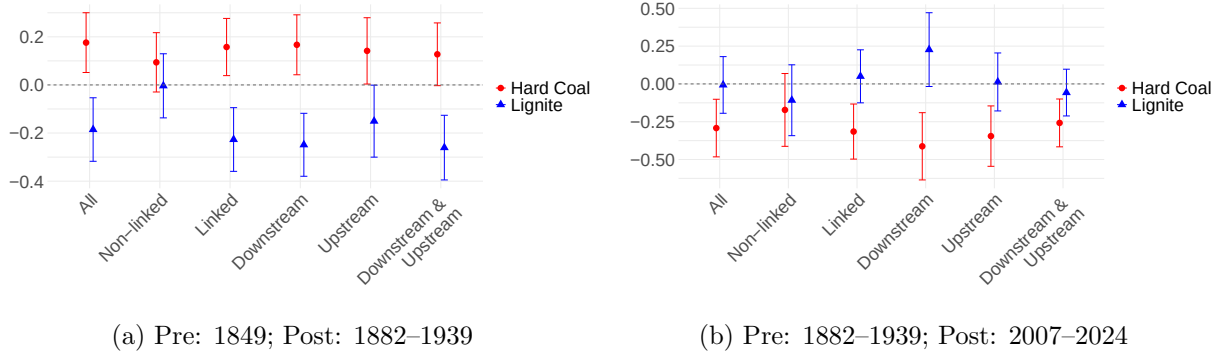


Figure A15: Difference-in-Differences regressions: Effect of Coal Endowment on Disaggregated Manufacturing Employment using German I-O table

Note: This figure plots coefficients and 95% confidence intervals from difference-in-differences regressions based on Equation 1, where the event-time indicators are replaced by a single post-period indicator. Both panels report estimated effects of hard coal (red) and lignite (blue) endowments on county-level manufacturing employment, disaggregated by linkage to coal mining using the German IO table from 1936 (Fremdling & Staeglin, 2014). All outcomes are log-transformed and standardized to have a mean of zero and a sd of one. The left panel covers the extraction period (1849–1939) using counties within Prussian borders. The post-period indicator equals one for all years after 1849. The right panel shows results for the coal phase-out period (1882–2024) in West Germany, where the post-period indicator equals one for the years 2007 to 2024. Standard errors are clustered at the county level.

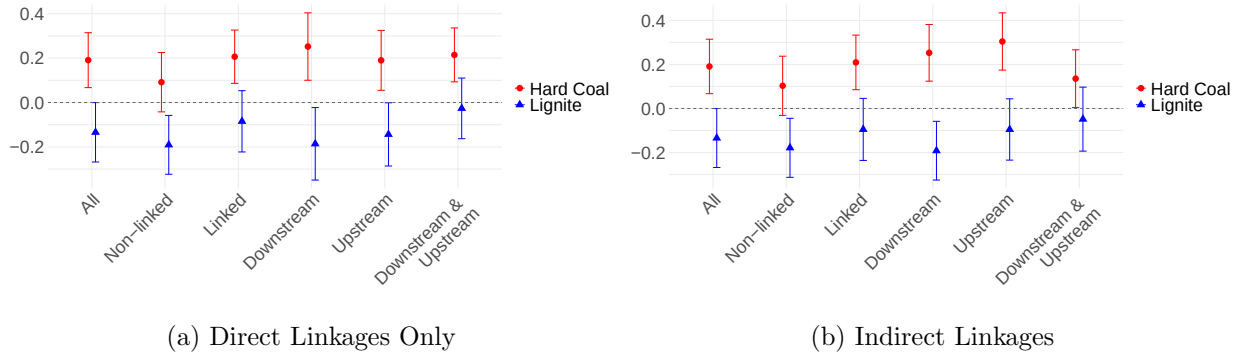


Figure A16: Difference-in-Differences Regressions: Alternative Linkage Definitions

Note: This figure plots coefficients and 95% confidence intervals from difference-in-differences regressions based on Equation 1, where the event-time indicators are replaced by a single post-period indicator. Both panels report the estimated effects of hard coal (red) and lignite (blue) endowments on county-level manufacturing employment, disaggregated by linkage to coal mining, based on the U.S. input-output table from 1947 (U.S. Bureau of Economic Analysis, 2019). All outcomes are log-transformed and standardized to have a mean of zero and a sd of one. The analysis covers the extraction period (1849–1939) and is restricted to counties within Prussian borders. The post-period indicator equals one for all years after 1849. The left panel uses direct linkages only; the right panel uses indirect linkages for both upstream and downstream relationships. Standard errors are clustered at the county level.

A.2.4 Shift-Share Approach

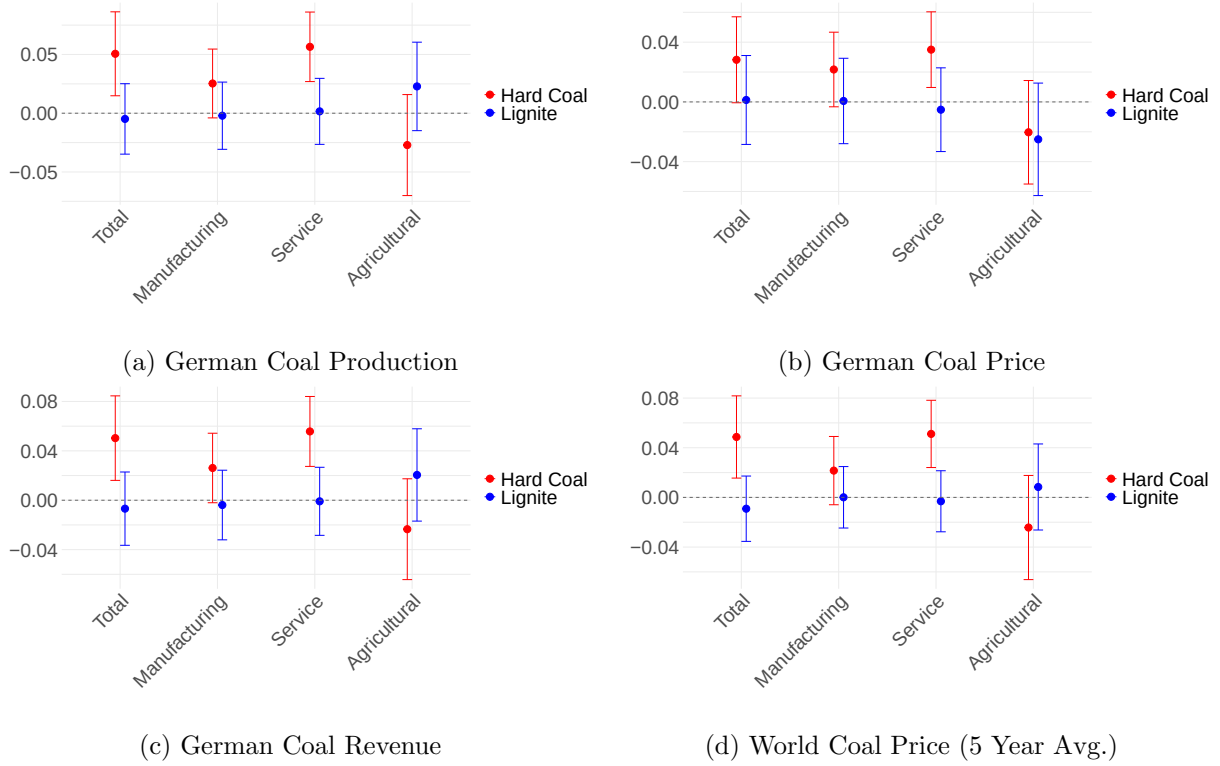
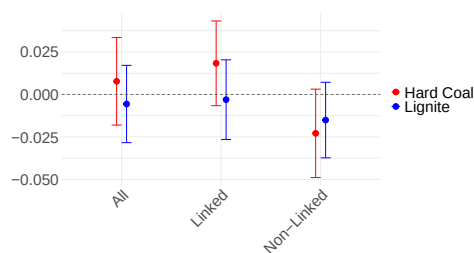
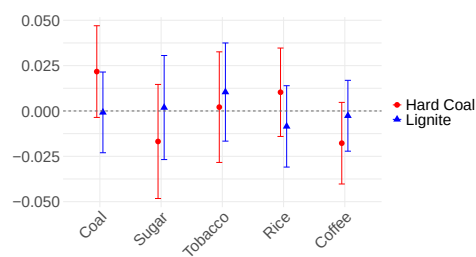


Figure A17: Reduced-Form Shift-Share Approach: Alternative Time-Series Variation

Note: This figure plots coefficients and 95% confidence intervals from reduced-form shift-share approach based on Equation 2, using alternative “shift” variables. Panel (a) uses German hard coal and lignite production; Panel (b) uses prices; Panel (c) uses revenue (price times quantity); and Panel (d) uses a 5-year moving average of global coal prices. Each estimate captures the differential effect of national hard coal (red) and lignite (blue) booms on counties with corresponding coal endowments, relative to non-endowed counties. The dependent variables are county-level employment in total, manufacturing, services, and agriculture. All outcomes are log-transformed and standardized to have a mean of zero and a sd of one. Standard errors are clustered at the county level. The analysis covers the period 1882–1939 within the borders of the German Empire.



(a) Disaggregated by Linkage



(b) Placebo Price Shocks

Figure A18: Reduced-Form Shift-Share Approach: Disaggregated Manufacturing Employment by Linkage

Note: This figure plots coefficients and 95% confidence intervals from reduced-form shift-share regressions based on Equation 2. Each estimate captures the differential effect of global price shocks on manufacturing employment. Panel (a) disaggregates manufacturing employment by linkage intensity to coal mining, showing responses to global coal price shocks in counties with hard coal and lignite endowments, relative to non-endowed counties. Panel (b) presents placebo estimates using global price shocks for unrelated commodities (sugar, tobacco, rice, coffee) to validate the identification strategy. All outcomes are log-transformed and standardized to mean zero and unit variance. Standard errors are clustered at the county level. The sample covers the period 1882–1939 within the borders of the German Empire.

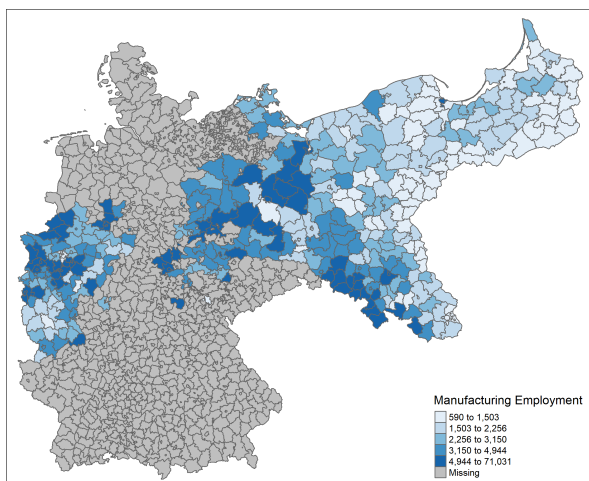
A.3 Spatial Coverage and Harmonization of Administrative Boundaries

Administrative county borders changed repeatedly over the course of my study period due to wars, shifts in sovereignty, and territorial-administrative reforms. To ensure consistency in spatial units across time, I harmonize all data to the geographic boundaries of counties as defined in 1900. Specifically, I overlay historical shapefiles from each census year onto the 1900 county map and aggregate data by weighting historical counties according to the proportion of their area overlapping with 1900 boundaries.

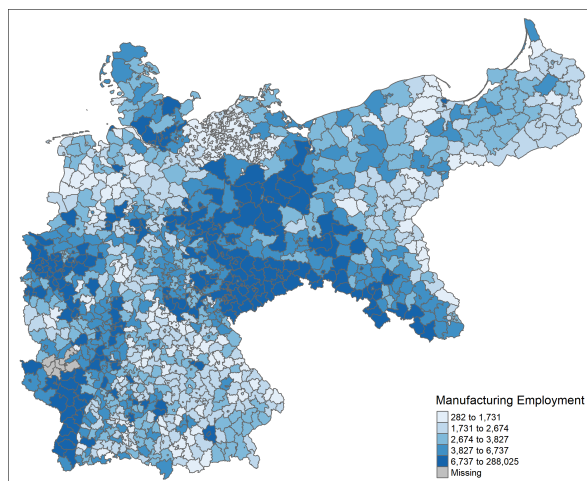
Territorial coverage also varies by year. For instance, the 1849 data include only the Kingdom of Prussia, which at the time encompassed large parts of what would later form the German Empire. In contrast, some eastern regions that were part of Germany in earlier decades, such as Silesia and East Prussia, lie outside present-day borders and are not covered in more recent census waves. To address this, I conduct separate analyses for periods with largely stable territorial coverage and provide robustness checks using balanced panels of counties observed throughout.

Figures A19 and A20 illustrate the geographic coverage of my dataset over time. Blue-shaded counties indicate regions for which manufacturing employment data are available in each year, while grey-shaded areas correspond to regions for which data are missing or which were not part of Germany at the time. These maps highlight how the availability of consistent county-level data varies across historical periods due to administrative boundary changes, wars, and political realignments.

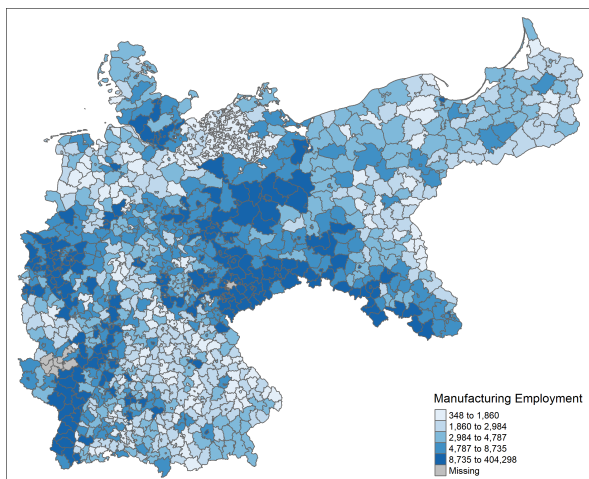
For the 1849–1939 period, I restrict the analysis to counties within Prussia, where the earliest employment data are available ($N = 417$). For the period 1882–1939, the sample includes counties within the borders of the German Empire ($N = 1122$). For the coal phase-out period (1950–2024), I focus on counties in West Germany (Federal Republic of Germany), excluding the former German Democratic Republic (GDR), which offer largely stable and continuous geographic coverage over time ($N = 640$).



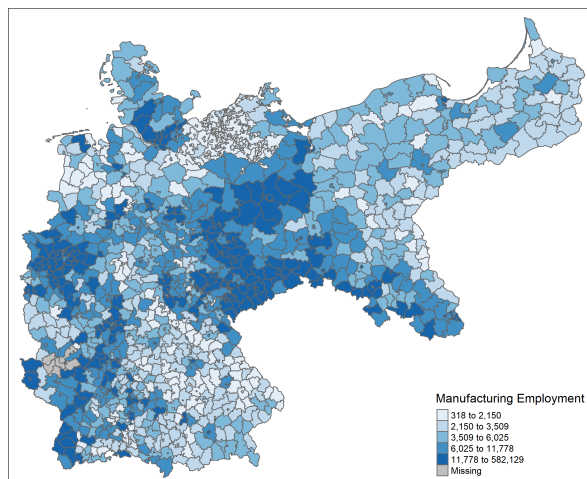
(a) 1849



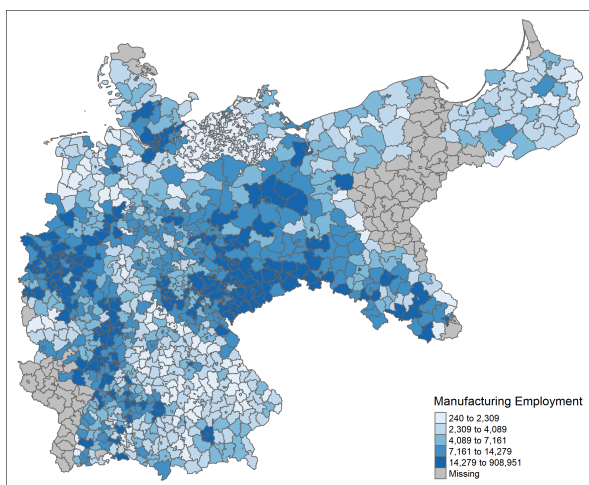
(b) 1882



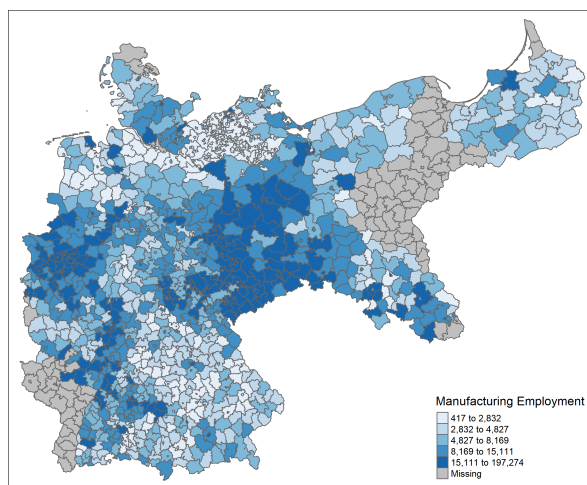
(c) 1895



(d) 1907

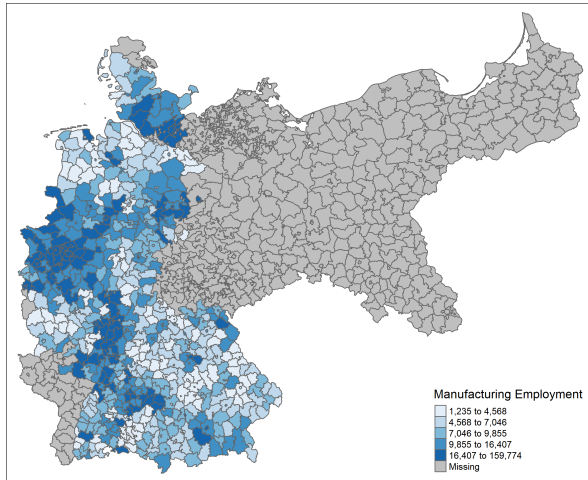


(e) 1925

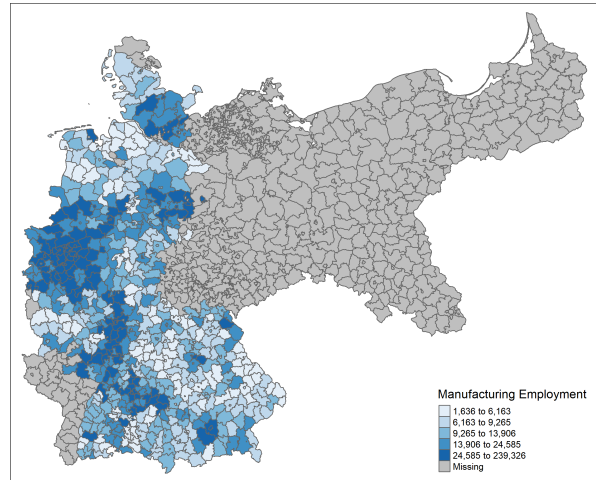


(f) 1939

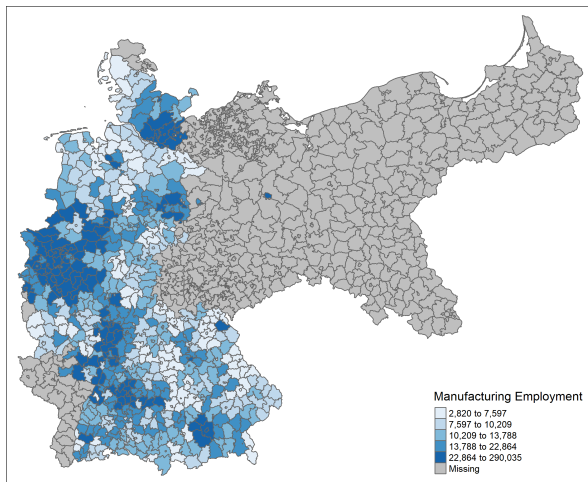
Figure A19: Manufacturing employment in German Counties (1849–1939)



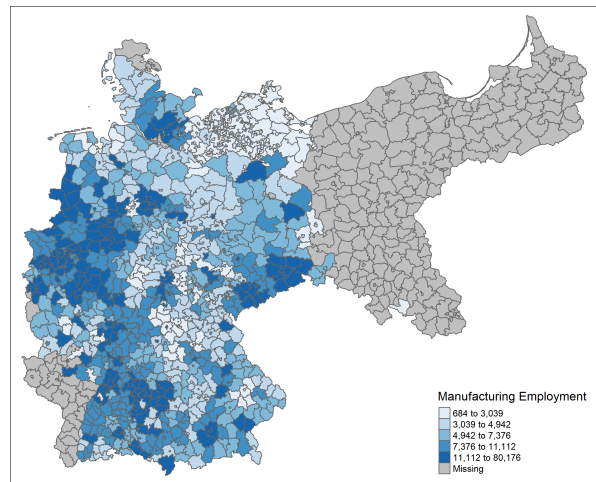
(a) 1950



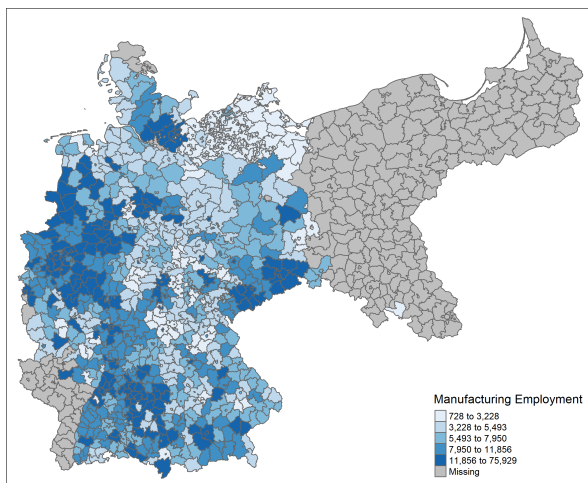
(b) 1961



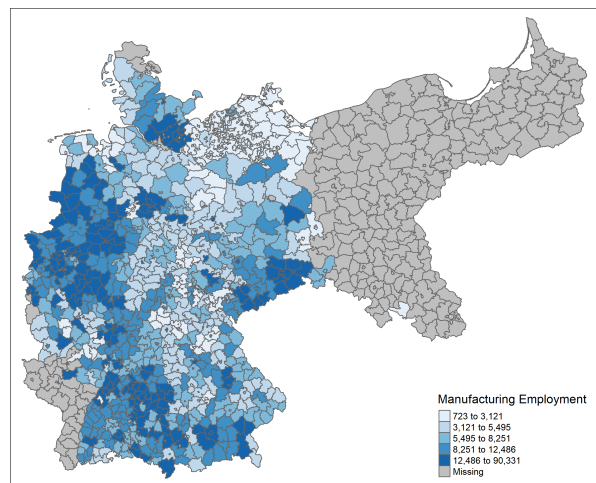
(c) 1987



(d) 2007



(e) 2015



(f) 2024

Figure A20: Manufacturing employment in German Counties (1950–2024)

A.4 List of Variables and Data Sources

A.4.1 Outcome Variables

Sectoral Employment 1849–2024: County-level employment data for 1849 is drawn from Statistisches Bureau zu Berlin (1851–1855). Data for the years 1882, 1895, 1907, 1925, 1933, and 1939 are taken from the respective occupation censuses in Kaiserliches Statistisches Amt, 1884, 1897, 1910; Statistisches Reichsamt, 1929, 1936b, 1943. Employment data for 1950, 1961, 1970, and 1987 are sourced from the Statistisches Bundesamt, 1953, 1964, 1972, 1989. Annual employment data from 2007 to 2024 are obtained from the Bundesagentur für Arbeit (2025).

Unskilled Wages 1800–1914: Average daily wage in Mark of unskilled male “seasonal fill” workers (day laborers) employed in public state forestry during the time periods 1800–1809, 1810–1819, 1820–1829, 1830–1839, 1840–1849, 1850–1859, 1860–1869, 1870–1874, and 1875–1879 as reported by Eggert (1883). Average local daily wages in Mark for manual day laborers in a county for the year 1883 as reported by Schmitz, 1888 and for the years 1892 to 1914 as reported by Zentralblatt, 1892. The complete digitized wage series is compiled by A.-K. Becker and Hornung (2025).

A.4.2 Treatment Variables

Hard Coal (Indicator): Binary variable that takes a value of one for counties with access to hard coal resources within Germany, defined as those with rock strata formed during the Carboniferous period (approximately 295 to 320 million years ago) within 20 km of the county’s geodesic centroid; zero otherwise. Data source: Asch and Bellenberg, 2005.

Lignite (Indicator): Binary variable that takes a value of one for counties with access to lignite resources within Germany, defined as those with rock strata formed during the Miocene, Oligocene, Paleogene, or Pliocene period (approximately 2 to 66 million years ago) where lignite is among the two most prevalent rock types within 20 km of the county’s geodesic centroid; zero otherwise. Data source: Asch and Bellenberg, 2005.

Hard Coal Field (Continuous): This variable measures the distance in kilometers from a county centroid to the nearest hard coal field. The underlying geospatial data on major nineteenth-century coalfields in Europe is taken from Châtel and Dollfus (1931) and digitized by Fernihough and O’Rourke (2021). Because the dataset includes both lignite and hard coal fields, I manually classified each site based on historical sources of mine locations, retaining only hard coal fields. The distance measure is truncated at zero, that is, if a county centroid lies directly on a hard coal field, the value is set to zero.

Lignite Field (Continuous): This variable measures the distance in kilometers from a county centroid to the nearest lignite field. The underlying geospatial data on major nineteenth-century

coalfields in Europe is taken from Châtel and Dollfus (1931) and digitized by Fernihough and O'Rourke (2021). Because the dataset includes both lignite and hard coal fields, I manually classified each site based on historical sources of mine locations, retaining only lignite fields. The distance measure is truncated at zero, that is, if a county centroid lies directly on a lignite coal field, the value is set to zero.

A.4.3 Shift Variables

Global Coal Prices 1882–1939: I use annual real world coal prices from Jacks (2019), expressed in 1900 U.S. dollars, for all years between 1882 and 1939.

Global Prices 1882–1939: I use annual real world prices for rice, tobacco, sugar, and coffee from Jacks (2019), expressed in 1900 U.S. dollars, for 1882–1939.

German Hard Coal and Lignite Production, Prices, and Revenues 1882–1939: Data on German hard-coal and lignite prices and production for 1882–1939 come from (Fischer, 2011). Revenues for each resource are computed as the product of the average national production quantity and the corresponding average national price in a given year.

A.4.4 Geographic Control Variables

Temperature: The average temperature (in degrees Celsius) in a county during the 1970–2000 period, constructed by temporally and spatially aggregating time series data on mean monthly temperature at a geospatial resolution of 30 arc seconds (i.e., grid cells of approximately 1 square kilometer), obtained from the *WorldClim (version 2.1) dataset* (Fick & Hijmans, 2017).

Rainfall: The average precipitation (in millimeters) in a county during the 1970–2000 period, constructed by temporally and spatially aggregating time series data on total monthly precipitation at a geospatial resolution of 30 arc seconds (i.e., grid cells of approximately 1 square kilometer), obtained from the *WorldClim (version 2.1) dataset* (Fick & Hijmans, 2017).

Slope: Average land slope (in degrees) within a county, derived from geospatial elevation data at a resolution of 30 arc seconds (approximately 1 km × 1 km). Slope is calculated using the eight nearest neighboring cells, based on the *WorldClim version 2.1 dataset* (Fick & Hijmans, 2017). This measure captures local terrain steepness, with higher values indicating steeper gradients.

Distance to Coast: The distance in kilometers from a county centroid to the nearest coast, constructed using a map of all European coastlines provided by the European Environment Agency (EEA) (<https://www.eea.europa.eu/data-and-maps/data/eea-coastline-for-analysis-1/gis-data/europe-coastline-shapefile>)

Distance to Navigable River: The distance in kilometers from a county centroid to the nearest navigable river, constructed using a map of waterways (Schiffahrtsstraßen) in the German Customs Union (Zollverein) in 1850, available through the “Server for Digital Historical Maps” at the Leibniz Institute of European History at the University of Mainz (IEG, 2010).

Caloric Suitability Index:

The average caloric suitability index in a county, as constructed by Galor and Özak, 2016. This index reflects potential agricultural productivity (measured in calories) in 5’x5’ grid cells, based on the crops available for cultivation.